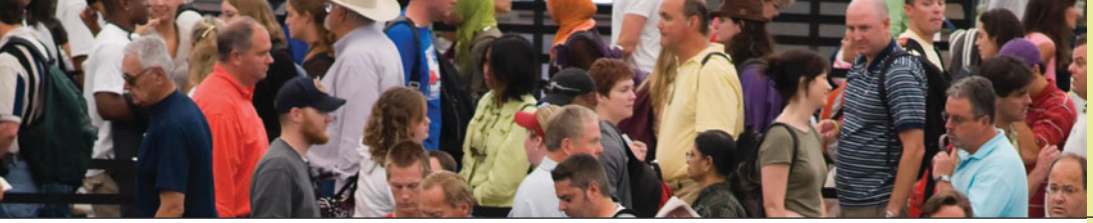
CHAPTER 2

Social Cognition

How We Think About the Social World





THE PROPOSAL TO BUILD A MOSQUE WITHIN AN ISLAMIC CULTURAL CENTER near Ground Zero in New York City created a lot of conflict. Those on the anti-mosque side are vehemently opposed to the mosque being built where the developers want to build it. These folks say that of course the mosque can be built anywhere that the law allows, but "sensitivities" call for it to be moved "further away."

On the other side, Mayor Michael Bloomberg has said that we cannot allow ourselves to be talked into the idea of moving the planned mosque's future location. He claims there is no justification for moving it—that the opposition has the wrong idea entirely. In his view, locating the mosque elsewhere means that the 9/11 terrorists have accomplished their goal of either cowing us into submission and/or making us fight among ourselves.

Perhaps a social psychological analysis of how people think about the social world can help us to deconstruct this conflict. As you will see in this chapter, people often use mental shortcuts or rules of thumb to arrive at judgments. One that people use a lot is called the *representativeness heuristic,* a rule of thumb wherein people judge a current event by considering how much it resembles another event or category. One of the key symptoms of judging by representativeness is called "ignoring the base rate." Let's see how this can help us understand the debate about the mosque placement in New York.

At the time of the 9/11 attack there were about 900 million peaceful Muslims in the world. We're talking about Arabs throughout the Middle East, but also Turkey, India, Indonesia, and parts of Africa. And, of course, that 900 million includes the 6 million Muslims living in the United States. As for Al-Qaeda's numbers, on ABC's "This Week" in June 2010, Leon Panetta (Director of the CIA) said that there are probably less than 50 Al-Qaeda members hiding out in Pakistan. But let's allow for the possibility of thousands more in Yemen, Somalia, Afghanistan, and other places in which Al-Qaeda could be hanging out. All told, let's speculate that our total complement of Al-Qaeda is 9,000 or less.

Given the overall population of Muslims in the world (900 million) and the Al-Qaeda number as 9,000, that would mean we have a ratio of 9 Al-Qaeda for every 900,000 Muslims, or, dividing by 9, about 1 Al-Qaeda member for every 100,000 peaceful Muslims. No matter how hard you try, it is quite ridiculous to make a judgment about 100,000 Muslims who have never attacked Americans based on the attitudes or actions of one member of Al-Qaeda. This is a clear example of ignoring the base rate.

CHAPTER OUTLINE

Heuristics: How We Reduce Our Effort in Social Cognition

- Representativeness: Judging by Resemblance
- Availability: "If I Can Retrieve Instances, They Must Be Frequent"
- Anchoring and Adjustment: Where You Begin Makes a Difference
- Status Quo Heuristic: "What Is, Is Good"

Schemas: Mental Frameworks for Organizing Social Information

- The Impact of Schemas on Social Cognition: Attention, Encoding, Retrieval
- Priming: Which Schemas Guide Our Thought?
- Schema Persistence: Why Even Discredited Schemas Can Sometimes Influence Our Thought and Behavior
- Reasoning by Metaphor: How Social Attitudes and Behavior Are Affected by Figures of Speech

Automatic and Controlled Processing: Two Basic Modes of Social Thought

- Automatic Processing and Automatic Social Behavior
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SOCIAL LIFE IN A CONNECTED WORLD
Dealing with Information Overload and Improving Choices

Potential Sources of Error in Social Cognition: Why Total Rationality Is Rarer Than You Think

- A Basic "Tilt" in Social Thought: Our Powerful Tendency to Be Overly Optimistic
- Situation-Specific Sources of Error in Social Cognition: Counterfactual Thinking and Magical Thinking

Affect and Cognition: How Feelings Shape Thought and Thought Shapes Feelings

- The Influence of Affect on Cognition
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EMOTIONS AND SOCIAL COGNITION
Why We Can't Always Predict Our Responses to Tragedy

- Affect and Cognition: Social Neuroscience Evidence for Two Separate Systems



FIGURE 2.1 Using the Representativeness Heuristic and Ignoring the Base Rate
As these protestors of building an Islamic Cultural Center including a mosque in New York imply on their signs, all Muslims are being judged in terms of their presumed resemblance to the 9/11 terrorism-perpetrators. Of course, the base rate of almost 1 billion Muslims in the world who live peacefully and do not commit nor support such crimes is ignored when the representativeness heuristic is employed.

But people might try anyway, so let's take up a second argument. Another aspect of the representative heuristic is the nature of that which is being represented. After 9/11, people's perceptions of Muslims changed. Before 9/11, Arab Muslims in particular were perhaps seen as backward desert-dwellers, but not as threatening or dangerous to Americans. But how representative is Al-Qaeda of the 900 million Muslims in the world? That is, if Al-Qaeda were the "army" of Muslims everywhere, then we might feel more justified in blaming all people of the Islamic faith for 9/11. But, in fact, across the Muslim world, Al-Qaeda is considered a deviant group. By deviant, we mean that the attitudes and beliefs, as well as the behaviors of Al-Qaeda, are markedly different from peaceful Muslims.

How so, you might ask? Well, for one thing, peaceful Muslims may get mad just as you and I do, but they do *not* believe that the Koran permits the indiscriminate killing of 3,000 innocent people, as was done on 9/11 by Al-Qaeda. Thus, the actions of Al-Qaeda are not representative of the general population of Muslims, and have almost nothing to do with the religion of Islam and the Koran as understood by ordinary Muslim devotees.

Of course, we use the representativeness heuristic every day as a shortcut to forming opinions about people in various groups and the probability that they will behave in particular ways. But, in the case of the so-called Ground Zero mosque, use of the representativeness heuristic as shown in Figure 2.1 alters people's perception of the blameworthiness of Islam with regard to 9/11, and that changes people's impressions of whether an Islamic place of worship should be built close to Ground Zero.



Building a mosque near Ground Zero . . . what, you may be wondering, does this have to do with the major focus of this chapter, **social cognition**—how we think about the social world, our attempts to understand it, and ourselves and our place in it (e.g., Fiske & Taylor, 2008; Higgins & Kruglanski, 1996)? The answer is simple: this conflict captures several key issues relating to social cognition that we examine in the rest of this chapter. First, it suggests very strongly that often our thinking about the social world proceeds on "automatic"—quickly, effortlessly, and without lots of careful reasoning. As we'll see later, such automatic thought or automatic processing offers important advantages—it requires

social cognition
The manner in which we interpret, analyze, remember, and use information about the social world.

little or no effort and can be very efficient. While such automatic processes, including heuristic use, can lead to satisfactory judgments, it can also lead to important errors in the conclusions we draw.

This incident also illustrates that although we do a lot of social thought on "automatic," we do sometimes stop and think much more carefully and logically about it (e.g., Should one Muslim's actions be taken as representative of 100,000 Muslims?). Such *controlled processing*, as social psychologists term it, tends to occur when something unexpected happens—something that jolts us out of automatic, effortless thought. For example, when New York's Mayor Bloomberg expressly questioned the validity of comparing "Muslims" to the 9/11 attackers, and argued that moving the mosque elsewhere would mean that the terrorists had won by making the United States a less free society, some people did indeed question their initial premise. As we'll see in later sections, unexpected events often trigger such careful, effortful thought.

In the remainder of this chapter, we examine the several types of **heuristics**—simple rules of thumb we often use to make inferences quickly, and with minimal effort—that people frequently use, and describe the research conducted by social psychologists addressing how they operate. Next, we consider in-depth the idea that often, social thought occurs in an automatic manner. In other words, it often unfolds in a quick and relatively effortless manner rather than in a careful, systematic, and effortful one. We consider how a basic component of social thought—*schemas*, or mental frameworks that allow us to organize large amounts of information in an efficient manner—can exert strong effects on social thought—effects that are not always beneficial from the point of view of accuracy. After considering how schema use can lead to judgment errors, we examine several specific tendencies or "tilts" in social thought—tendencies that can lead us to false conclusions about others or the social world. Finally, we focus on the complex interplay between **affect**—our current feelings or moods—and various aspects of social cognition (e.g., Forgas, 1995a, 2000).

heuristics
Simple rules for making complex decisions or drawing inferences in a rapid manner and seemingly effortless manner.

affect
Our current feelings and moods.

Heuristics: How We Reduce Our Effort in Social Cognition

Several states have passed or are considering adopting laws that ban talking on hand-held cell phones and texting while driving. Why? Because—as the cartoon in Figure 2.2 indicates—these are very dangerous practices, particularly texting. It has been found over and over again that when drivers are distracted, they are more likely to get into accidents, and talking or texting can certainly be highly distracting. What about global positioning systems (GPS), which show maps to drivers; do you think that they, too, can lead to distraction and cause accidents?

At any given time, we are capable of handling a certain amount of



FIGURE 2.2 Distraction: A Potential Cause of Accidents
Our capacity to process incoming information is definitely limited, and can easily be exceeded. This can happen when drivers are texting or talking on the phone while driving. As this cartoon suggests, fatal accidents can result.

information; additional input beyond this puts us into a state of **information overload** where the demands on our cognitive system are greater than its capacity. In addition, our processing capacity can be depleted by high levels of stress or other demands (e.g., Chajut & Algom, 2003). To deal with such situations, people adopt various strategies designed to "stretch" their cognitive resources—to let them do more, with less effort, than would otherwise be the case. This is one major reason why so much of our social thought occurs on "automatic"—in a quick and effortless way. We discuss the costs and potential benefits of such thought later. Here, however, we focus on techniques we use to deal quickly with large amounts of information, especially under **conditions of uncertainty**—where the "correct" answer is difficult to know or would take a great deal of effort to determine. While many strategies for making sense of complex information exist, one of the most useful tactics involves the use of *heuristics*—simple rules for making complex decisions or drawing inferences in a rapid and efficient manner.

Representativeness: Judging by Resemblance

Suppose that you have just met your next-door neighbor for the first time. While chatting with her, you notice that she is dressed conservatively, is neat in her personal habits, has a very large library in her home, and seems to be very gentle and a little shy. Later you realize that she never mentioned what she does for a living. Is she a business manager, a physician, a waitress, an artist, a dancer, or a librarian? One quick way of making a guess is to compare her with your **prototype**—consisting of the attributes possessed by other members of each of these occupations. How well does she resemble people you have met in each of these fields or, perhaps, the typical member of these fields (Shah & Oppenheimer, 2009)? If you proceed in this manner, you may quickly conclude that she is probably a librarian; her traits seem closer to those associated with this profession than they do to the traits associated with being a physician, dancer, or executive. If you made your judgment about your neighbor's occupation in this manner, you would be using the **representativeness heuristic.** In other words, you would make your judgment on the basis of a relatively simple rule: The more an individual seems to resemble or match a given group, the more likely she or he is to belong to that group.

Are such judgments accurate? Often they are, because belonging to certain groups does affect the behavior and style of people in them, and because people with certain traits are attracted to particular groups in the first place. But sometimes, judgments based on representativeness are wrong, mainly for the following reason: Decisions or judgments made on the basis of this rule tend to ignore *base rates*—the frequency with which given events or patterns (e.g., occupations) occur in the total population (Kahneman & Frederick, 2002; Kahneman & Tversky, 1973). In fact, there are many more business managers than librarians—perhaps 50 times as many. Thus, even though your neighbor seemed more similar to the prototype of librarians than managers in terms of her traits, the chances are actually higher that she is a manager than a librarian. Likewise, as we saw in the opening example, ignoring the base rate that consists of millions of Muslims who are nonviolent can lead to errors in our thinking about people.

The representativeness heuristic is used not only in judging the similarity of people to a category prototype, but also when judging whether specific causes resemble and are therefore likely to produce effects that are similar in terms of magnitude. That is, when people are asked to judge the likelihood that a particular effect (e.g., either many or a few people die of a disease) was produced by a particular cause (e.g., an unusually infectious bacteria or a standard strain), they are likely to expect the strength of the cause to match its effect. However, cultural groups differ in the extent to which they rely on the representative heuristic and expect "like to go with like" in terms of causes and effects. In particular, people from Asia tend to consider more potential causal factors when judging effects than do Americans (Choi, Dalal, Kim-Prieto, & Park, 2003). Because they consider more information and arrive at more complex attributions when judging an event,

information overload
Instances in which our ability to process information is exceeded.

conditions of uncertainty
Where the "correct" answer is difficult to know or would take a great deal of effort to determine.

prototype
Summary of the common attributes possessed by members of a category.

representativeness heuristic
A strategy for making judgments based on the extent to which current stimuli or events resemble other stimuli or categories.

Asians should show less evidence of thinking based on the representative heuristic—a judgment simplification strategy—compared to North Americans.

To test this reasoning, Spina et al. (2010) asked students in China and Canada to rate the likelihood that a high- or low-magnitude effect (few or many deaths) was caused by a virus that differed in magnitude (a strain that was treatment-resistant or a standard strain that could be controlled with medical treatment). While participants in both national groups showed evidence of expecting high-magnitude effects (many deaths) to be produced by high-magnitude causes (the treatment-resistant virus strain) and low-magnitude effects (few deaths) to be produced by low-magnitude causes (the standard strain of the virus), Canadian participants showed this effect much more strongly than the Chinese participants. Such reasoning differences could potentially result in difficulty when members of different groups seek to achieve agreement on how best to tackle problems affecting the world as a whole—such as climate change. Westerners may expect that "big causes" have to be tackled to reduce the likelihood of global warming, whereas Asians may be comfortable emphasizing more "minor causes" of substantial outcomes such as climate change.

Availability: "If I Can Retrieve Instances, They Must Be Frequent"

When estimating event frequencies or their likelihood, people may simply not know the "correct" answer—even for events in their own lives. So how do they arrive at a response? Ask yourself, how often have you talked on your cell phone while driving? Well, *I* can remember quite a few instances, so I'd have to guess it is quite often. This is an instance of judging frequency based on the ease with which instances can be brought to mind. Now consider another, non-self-related question: Are you safer driving in a huge SUV or in a smaller, lighter car? Many people would answer: "In the big SUV"—thinking, as shown in Figure 2.3, that if you are in an accident, you are less likely to get hurt in a big vehicle compared to a small one. While that might seem to be correct, actual data indicate that death rates (number of deaths per 1 million vehicles on the road) are higher for SUVs than smaller cars (e.g., Gladwell, 2005). So why do so many people conclude, falsely, that they are safer in a bulky SUV? Like the cell phone–use question, the answer seems to involve what comes to mind when we think about this question. Most people can recall scenes in which a huge vehicle had literally crushed another smaller vehicle in an accident. Because such scenes are dramatic, we can readily bring them to mind. But this "ease of retrieval" effect may mislead us: We assume that because such scenes are readily available in memory, they accurately reflect the overall frequency, when, in fact, they don't. For instance, such recall does not remind us of the fact that SUVs are involved in accidents more often than smaller, lighter cars; that large SUVs tip over more easily than other vehicles; or



FIGURE 2.3 Availability Heuristic Use: Images Like These Come Readily to Mind

People believe they are safer and less likely to get into an accident with a larger SUV than a smaller car—in part, because images like these come readily to mind. But, actually, SUVs are involved in more accidents than smaller cars.

that SUVs are favored by less careful drivers who are more likely to be involved in accidents!

This and many similar judgment errors illustrate the operation of the **availability heuristic,** another cognitive "rule of thumb" suggesting that the easier it is to bring information to mind, the greater its impact on subsequent judgments or decisions. While use of this heuristic can make good sense much of the time—after all, the fact that we can bring some types of information to mind quite easily suggests that it may indeed be frequent or important so it *should* influence our judgments and decisions. But relying on availability in making social judgments can also lead to errors. Specifically, it can lead us to overestimate the likelihood of events that are dramatic but rare because they are easy to bring to mind. Consistent with this principle, many people fear travel in airplanes more than travel in automobiles, even though the chances of dying in an auto accident are hundreds of times higher. Likewise, people overestimate murder as a cause of death, and underestimate more mundane but much more frequent killers such as heart disease and stroke. The idea here is that because of the frequency that murder and other dramatic causes of death are presented in the mass media, instances are easier to retrieve from memory than are various natural causes of death that are rarely presented in the media. Here's another example: Physicians who examine the same patient often reach different diagnoses about the patient's illness. Why? One reason is that physicians have different experiences in their medical practices, and so find different kinds of diseases easier to bring to mind. Their diagnoses then reflect these differences in ease of retrieval—or, their reliance on the availability heuristic.

Interestingly, research suggests that there is more to the availability heuristic than merely the subjective ease with which relevant information comes to mind. In addition, the amount of information we can bring to mind seems to matter, too (e.g., Schwarz et al., 1991). The more information we can think of, the greater its impact on our judgments. Which of these two factors is more important? The answer appears to involve the kind of judgment we are making. If it is one involving emotions or feelings, we tend to rely on the "ease" rule, whereas if it is one involving facts or the task is inherently difficult, we tend to rely more on the "amount" rule (e.g., Rothman & Hardin, 1997; Ruder & Bless, 2003).

It is also the case that the ease of bringing instances to mind affects judgments that are self-relevant more readily than judgments about others. In fact, even judgments about objects that we are personally familiar with—say, consumer brands—are influenced by ease of retrieval more than judgments about brands that we are less familiar with (Tybout, Sternthal, Malaviya, Bakamitsos, & Park, 2005). This is because when we are aware that we have less information about others or unfamiliar objects, making judgments about them seems more difficult and ease of retrieval is given less weight. But when we think we are familiar with the task, know more about the task, or the task itself is easy, then ease of retrieval is particularly likely to be the basis of our judgment. Let's see how this plays out in judgments of risk.

Harvard University students were asked to make judgments about how safe their college town, Cambridge, Massachusetts, was after they had been asked to recall either two or six examples of when they or another student "had felt unsafe or feared for their safety around campus" (Caruso, 2008). Of course, it should be (and was for these participants) easier to recall two instances when they felt unsafe than to recall six instances, and it should be easier to retrieve instances when you felt a particular way than when another person did. Those students who had an easy job of recalling unsafe examples for themselves rated their town as more unsafe than when they had a difficult time retrieving more examples. Use of the perceived ease of recall, though, was not applied to judgments of the safety of one's own town when the examples brought to mind concerned someone else's experiences. Consider another example: Would you find it easier to generate two instances that are diagnostic of your creativity, or six instances? What about instances for an acquaintance? As shown in Figure 2.4, students did find it easier to generate

availability heuristic
A strategy for making judgments on the basis of how easily specific kinds of information can be brought to mind.

two examples of their own creativity compared to six examples, and this influenced their ratings of their own creativity. Ease of retrieving examples of creativity for an acquaintance did not affect ratings of creativity for that other because subjective ease of retrieval is given less weight.

Anchoring and Adjustment: Where You Begin Makes a Difference

When people attempt to sell something—whether it be a house on HGTV, or a car through an ad in a newspaper—they typically set the "asking" price higher than they really expect to get. Likewise, buyers often bid initially less than they expect to ultimately pay. This is mostly because buyers and sellers want to give themselves some room for bargaining. Often the selling price is the starting point for discussion; the buyer offers less, the seller counters, and the process continues until an agreement is reached, or one or the other gives up. It turns out that when a seller sets a starting price, this is an important advantage because of another heuristic that strongly influences our thinking: **anchoring and adjustment.** This heuristic involves the tendency to deal with uncertainty in many situations by using something we do know as a starting point, and then making adjustments to it. The seller's price provides such a starting point, to which buyers try to make adjustments in order to lower the price they pay. Such lowering makes the buyer feel that, by comparison to the original asking price, they are getting a very good deal. This too is how "sale pricing" and highly visible "reductions" work in retail stores—the original starting point sets the comparison so shoppers feel like they are then getting a bargain.

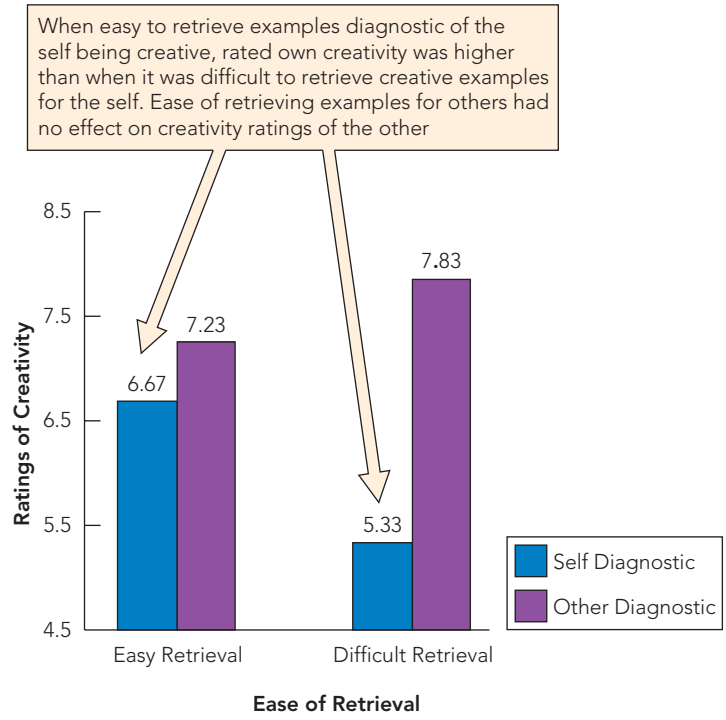


FIGURE 2.4 Availability Heuristic Use: Perceived Creativity of the Self Depends on Ease of Retrieval

Ratings of perceived self-creativity depended on ease of retrieval. When it was easy (vs. difficult) to generate diagnostic examples for the self, then perceived self-creativity increased. The ease or difficulty of generating creative instances for another person did not affect judgments of the other's creativity. (Source: Based on research by Caruso, 2008).

In a sense, the existence of the anchoring and adjustment heuristic is far from surprising. In uncertain situations we have to start somewhere. What is more surprising, however, is how powerful this effect is even in situations where, rationally, it should not operate. For instance, consider an unsettling study by Englich, Mussweiler, and Strack (2006), indicating that even court decisions and sentences can be strongly influenced by anchoring and adjustment and that, moreover, this occurs even for experienced judges!

In this research, the participants were highly experienced legal professionals in Germany. They were asked to read a very realistic court case and then learned of prison sentences recommended for the defendant. In one study, these recommendations were from a journalist—someone with no legal training. In another study, the recommended sentences were actually generated by throwing dice—randomly, and with no connection to the crime itself. Finally, in another, they were from an experienced prosecutor. Some of the recommendations were lenient (e.g., 1 month of probation) and others were harsh (e.g., 3 years in prison for the same crime). After receiving this information, the experienced legal participants made their own sentencing recommendations. The recommendations of these experts should *not* be influenced by the anchors they received, especially when the sources were either irrelevant or purely random in two conditions (lenient or

anchoring and adjustment heuristic
A heuristic that involves the tendency to use a number of value as a starting point to which we then make adjustments.

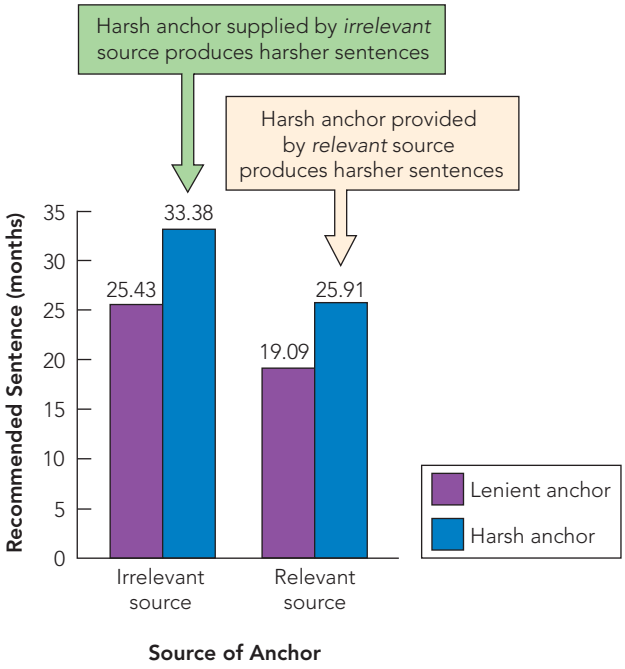


FIGURE 2.5 Anchoring and Adjustment in Legal Decisions

When experienced legal experts learned of the sentences recommended by an irrelevant source (someone with no legal training—a journalist, or even just a throw of a dice), their own recommendations were strongly influenced by these anchors. Harsher sentences were recommended when the anchors were harsh, and more lenient sentences when the anchors were lenient. The same anchoring effects were found when the source of the anchor was relevant—an experienced prosecutor. These findings indicate that anchoring often exerts powerful effects on social thought. (Source: Based on data from Englich, Mussweiler, & Strack, 2006).

harsh recommendations from a journalist or ones generated by the throw of dice). But, as you can see in Figure 2.5, these anchors did have significant effects: Sentences were harsher when participants were exposed to a harsh anchor but more lenient when they were exposed to a lenient anchor. Furthermore, it did *not* matter whether the source of the anchor was a journalist, an experienced prosecutor, or merely the throw of dice. These findings, while a compelling demonstration of the power of anchoring, are also quite disturbing. If even experienced and highly trained legal experts can be influenced by anchoring and adjustment, it seems clear that this is indeed a very powerful effect—and indicative of how shortcuts in social thought can have real consequences in important life contexts.

Why are the effects of the anchoring and adjustment heuristic so powerful? Research findings indicate that one reason is that although we do make adjustments to anchors, these adjustments are often not sufficient to overcome the initial impact of the anchors. In other words, we seem to stop as soon as a value we consider plausible is reached (Epley & Gilovich, 2006). In a sense, this is yet another example of the "save mental effort" principle that we tend to follow in many contexts and across many different aspects of social thought. Interestingly, the tendency to make insufficient judgments is greater when individuals are in a state in which they are less capable of engaging in effortful thought—for instance, after consuming alcohol or when people are busy doing other tasks (Epley & Gilovich, 2006). Overall, then, it appears that our tendency to let initial anchors influence our judgments—even in important situations—does stem, to an important degree, from a tendency to avoid the effortful work involved in making adjustments away from initial anchors.

Status Quo Heuristic: "What Is, Is Good"

When people are asked to make judgments and choices, they seem to act as though they believe the status quo is good. Similar to the availability heuristic, objects and options that are more easily retrieved from memory may be judged in a heuristic fashion as "good," as better than objects and options that are new, rarely encountered, or represent a change from the status quo. As with the other types of heuristics we've discussed, assuming that a product that has long been on the market is superior to a new version might seem to be logical because across time bad products tend to be removed from the market. But, it is also the case that old products stay on the market through inertia, and people may continue buying it partly out of habit. Indeed, many marketers seem to believe that people prefer new over the old—if their emphasis on "new and improved" on packaging is any indication!

In a series of studies, Eidelman, Pattershall and Crandall (2010) have put the issue of whether people heuristically favor "old" over "new," or the opposite, to the test. Participants in one study were given a piece of chocolate to taste. Before doing so, they were told either that the chocolate was first sold in its region of Europe in 1937 or in 2003. In the former case, the product was said to be on the market for 70 years and in the latter for only 3 years. Participants were then asked to rate how much they enjoyed the taste of the chocolate, whether they were impressed by it, and whether they would purchase it. They were then asked about the reasons for their evaluation of the chocolate. Overwhelmingly,

participants rated the chocolate that was said to have been in existence longer as more delicious than the chocolate that represented a new brand. These participants seemed to be unaware that time on the market had influenced their evaluations of the chocolate—they uniformly rated that as the least important reason for their evaluation and, instead, rated "its taste" as the most important factor affecting their evaluation. But, it was exactly the same chocolate and only the supposed length of time on the market differed! These researchers also showed in another experiment that students favored a degree requirement proposal that was said to already be in existence over the same proposal when it was framed as representing a change from the present. Furthermore, when the length of time a practice (acupuncture) was said to be in existence was varied—250, 500, 1,000, or 2,000 years—its perceived effectiveness increased across the time intervals. Likewise, a painting whose aesthetic qualities were to be judged was rated more pleasing when it was said to have been painted in 1905 compared to when it was said to have been painted more recently, in 2005. So, people do seem to use heuristically the length of time a product or practice has been in existence as a cue to its goodness. Although judgments of all products are unlikely to be biased in favor of age, and occasionally novelty may win, tradition or longevity often does seem to imply heuristically that the "tried and true" is better than the new.



KEY POINTS

- Because we have limited cognitive capacity, we often attempt to reduce the effort we expend on **social cognition**—how we think about other people and events. Given our limited capacity to process information, we often experience **information overload**. To deal with complex information, where the correct answer is not obvious **(conditions of uncertainty**), we make use of **heuristics**—simple rules for making decisions in a quick and relatively effortless manner.
- One such heuristic is **representativeness**, which suggests that the more similar an individual or subgroup of people is to typical members of a given group—the group's **prototype**—the more likely they will be seen as belonging to that group.
- Using the representativeness heuristic can lead to erroneous decisions when *base rates* are underused but are relevant.
- There are cultural differences in using representativeness to evaluate the likelihood that a particular cause was responsible for an effect. Asians tend to expect that "like will go with like" less than Westerners do.
- Another heuristic is **availability,** which suggests that the easier it is to bring information to mind, the greater its impact on subsequent decisions or judgments. In some cases, availability may also involve the amount of information we bring to mind. We tend to apply the ease of retrieval rule to judgments about ourselves more than to judgments about others.
- A third heuristic is **anchoring and adjustment**, which leads us to use a number or value as a starting point from which we then make adjustments. These adjustments may not be sufficient to reflect actual social reality, perhaps because once we attain a plausible value, we stop the process.
- Objects and options that are more *easily retrieved from memory* may be judged in a heuristic fashion as "good," as better than objects and options that are new, rarely encountered, or represent a change from the status quo.

Schemas: Mental Frameworks for Organizing Social Information

What happens when you visit your doctor? We all know it goes something like this. You enter and give your health insurance information. Then you sit and wait! If you are lucky, the wait is not very long and a nurse takes you into an examining room. Once there, you wait some more. Eventually, the doctor enters and talks to you and perhaps examines you. Finally, you leave and perhaps pay some part of your bill (the co-pay) on the way out. It doesn't matter who your doctor is or where you live—this sequence of events,





FIGURE 2.6 Schemas: Mental Frameworks Concerning Routine Events
Through experience, we acquire schemas—mental frameworks for organizing, interpreting, and processing social information. For instance, you almost certainly have well-developed schemas for such events as boarding an airplane (top photo) and going to the dentist (bottom photo). In other words, you know what to expect in these and many other situations, and are prepared to behave in them in certain sequences.

or something very much like it, will take place. None of this surprises you; in fact, you expect this sequence to occur—including the waiting. Why? Through past experience, you have built up a mental framework containing the essential features of this kind of situation—visiting a health professional. Similarly, you have formed other mental frameworks reflecting going to restaurants, getting a haircut, shopping for groceries, going to the movies, or boarding an airplane (see Figure 2.6).

Social psychologists term such frameworks **schemas,** and define them as mental frameworks that help us to organize social information, and that guide our actions and the processing of information relevant to those contexts. Since your personal experience in such situations is probably similar to that of others in your culture, everyone in a given society will tend to share many basic schemas. Once schemas are formed, they play a role in determining what we notice about the social world, what information we remember, and how we use and interpret such information. Let's take a closer look at these effects because as we'll soon see, they exert an important impact on our understanding of the social world and our relations with other people.

The Impact of Schemas on Social Cognition: Attention, Encoding, Retrieval

How do schemas influence social thought? Research findings suggest that they influence three basic processes: attention, encoding, and retrieval. *Attention* refers to what information we notice. *Encoding* refers to the processes through which information we notice gets stored in memory. Finally, *retrieval* refers to the processes through which we recover information from memory in order to use it in some manner—for example, in making judgments about other people.

Schemas have been found to influence all of these aspects of social cognition (Wyer & Srull, 1994). With respect to attention, schemas often act as a kind of filter: information consistent with them is more likely to be noticed and to enter our consciousness. Schemas are particularly likely to be relied on when we are experiencing *cognitive load*—when we are trying to handle a lot of information at one time (Kunda, 1999). In this case, we rely on our schemas because they help us process information efficiently.

Turning to encoding—the information that becomes the focus of our attention is much more likely to be stored in long-term memory. In general, it is information that is consistent with our schemas that is encoded. However, information that is sharply inconsistent with our schemas—information that does *not* agree with our expectations in a given situation—may be encoded into a separate memory location and marked with a unique "tag." Schema-inconsistent information is sometimes so unexpected that it literally seizes our attention and almost forces us to make a mental note of it (Stangor & McMillan, 1992). Here's an example: You have a well-developed schema for the role of "professor." You expect professors to come to class, to lecture, to answer questions, to give and grade exams, and so on. Suppose that one of your professors comes to class and instead of lecturing does magic tricks. You will certainly remember this experience because it is so inconsistent with your schema for professors—your mental framework for how professors behave in the classroom.

That leads us to the third process: retrieval from memory. What information is most readily remembered—information that is consistent with our schemas or information that is inconsistent with these mental frameworks? This is a complex question that has been investigated in many different studies (e.g., Stangor & McMillan, 1992; Tice, Bratslavky, & Baumeister, 2000). Overall, research suggests that people tend to report remembering

schemas
Mental frameworks centering on a specific theme that help us to organize social information.

information that is consistent with schemas more than information that is inconsistent. However, this could potentially stem from differences in actual memory or, alternatively, from simple response tendencies. In other words, information inconsistent with schemas might be present in memory as strongly as information consistent with schemas, but people simply report the information that is consistent with their schemas. In fact, the latter appears to be the case. When measures of memory are corrected for this response tendency, or when individuals are asked to actually *recall* information rather than indicate whether they recognize it, a strong tendency to remember information that is *incongruent* (i.e., does not fit) with schemas appears. So, the answer to the question, Which do we remember better—information consistent or inconsistent with our schemas?, *depends on the memory measure employed.* In general, people *report* information consistent with their schemas, but information inconsistent with schemas may be strongly present in memory, too.

Priming: Which Schemas Guide Our Thought?

We all develop a large array of schemas—cognitive frameworks that help us interpret and use social information. That raises an interesting question: Which of these frameworks influence our thought at any given point in time? One answer involves the strength of various schemas: the stronger and better-developed schemas are, the more likely they are to influence our thinking, and especially our memory for social information (e.g., Stangor & McMillan, 1992; Tice et al., 2000).

Second, schemas can be temporarily activated by what is known as **priming**—transitory increases in the ease with which specific schemas can be activated (Sparrow & Wegner, 2006). For instance, suppose you have just seen a violent movie. Now, you are looking for a parking spot and you notice one, but another driver turns in front of you and takes it first. Do you perceive her behavior as aggressive? Because the violent movie has activated your schema for "aggression," you may, in fact, be more likely to perceive her taking the parking spot as aggressive. This illustrates the effects of priming—recent experiences make some schemas active, and as a result, they exert effects on our current thinking.

Can priming be deactivated, or are we doomed to see the world in terms of the schema activated by our most recent experience? Social psychologists describe **unpriming** as a process by which thoughts or actions that have been primed by a recent experience dissipates once it finds expression. Unpriming effects are clearly demonstrated in a study by Sparrow and Wegner (2006). Participants were given a series of very easy "yes–no" questions (e.g., "Does a triangle have three sides?"). One group of participants was told to try to answer the questions randomly—*not* correctly. Another group responded to the questions twice; the first time, they were told to try to answer them correctly, while the second time, they were to try to answer them randomly. It was predicted that participants in the first group would not be able to answer the questions randomly; their schema for "answering correctly" would be activated, and lead them to provide the correct answers. In contrast, participants who answered the questions twice—first correctly and then randomly—would do better at responding randomly. Their first set of answers would provide expression for the schema "answer questions correctly," and so permit them to answer randomly the second time around. That's precisely what happened; those who only answered the question once and were told to do so randomly were actually correct 58 percent of the time—their activated schema prevented them from replying in a truly random manner. The participants who first answered the questions correctly and then randomly did much better: their answers the second time were correct only 49 percent of the time—they did show random performance. These findings indicate that once primed schemas are somehow expressed, unpriming occurs, and the influence of the primed schemas disappears. Figure 2.7 summarizes the nature of unpriming. If primed schemas are *not* expressed, however, their effects may persist for long periods of time—even years (Budson & Price, 2005; Mitchell, 2006).

priming
A situtation that occurs when stimuli or events increase the availability in memory or consciousness of specific types of information held in memory.

unpriming
Refers to the fact that the effects of the schemas tend to persist until they are somehow expressed in thought or behavior and only then do their effects decrease.

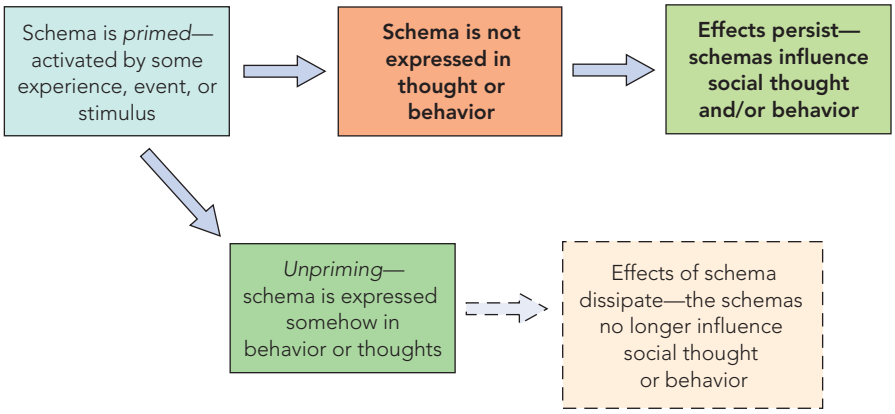


FIGURE 2.7 Unpriming of Schemas: Bringing the Effects of Priming to an End
When schemas are primed—activated by experiences, events, or stimuli, their effects tend to persist. In fact, they have been observed over years even. If the schema is somehow expressed in thought or behavior, however, unpriming may occur, and the impact of the schema may decrease or even disappear. (Source: Based on findings reported by Sparrow & Wegner, 2006).

Schema Persistence: Why Even Discredited Schemas Can Sometimes Influence Our Thought and Behavior

Although schemas are based on our past experience and are often helpful—they permit us to make sense out of a vast array of social information—they have an important "downside" too. By influencing what we notice, enter into memory, and later remember, schemas can produce distortions in our understanding of the social world. Unfortunately, schemas are often resistant to change—they show a strong **perseverance effect,** remaining unchanged even in the face of contradictory information (Kunda & Oleson, 1995). Perhaps even worse, schemas can sometimes be *self-fulfilling:* They influence our responses to the social world in ways that *make* it consistent with the schema!

Do our cognitive frameworks—our schemas—actually shape the social world as well as reflect it? A large body of evidence suggests that this is definitely so (e.g., Madon, Jussim, & Eccles, 1997; Smith, Jussim, & Eccles, 1999). Perhaps the most dramatic evidence that schemas can be self-fulfilling was provided by Rosenthal and Jacobson (1968), in a famous study of teachers and the unintended effects of their expectations on students. These researchers went to an elementary school and administered an IQ test to all students. Then they told the teachers that some of the students had scored very high and were about to "bloom" academically. The teachers were not given such information about other students, who constituted a control group. Although the researchers had chosen the names of the students for each group randomly, they predicted that this information would alter teachers' expectations about the children and their behavior toward them.

To find out if this was true, 8 months later the researchers tested both groups of children once again. Results were clear: those who had been described as "bloomers" to their teachers showed significantly larger gains on the IQ test than those in the control group. In short, teachers' beliefs about the students had operated in a *self-fulfilling manner:* The students whose teachers believed they would "bloom," actually did. So schemas can be a two-edged sword: They can help us make sense of the social world and process information efficiently, but they can also lock us into acting in ways that create the world that we expect.

Reasoning by Metaphor: How Social Attitudes and Behavior Are Affected by Figures of Speech

perseverance effect
The tendency for beliefs and schemas to remain unchanged even in the face of contradictory information.

metaphor
A linguistic device that relates or draws a comparison between one abstract concept and another dissimilar concept.

Might **metaphors**—linguistic devices that relate a typically abstract concept to another dissimilar concept—shape how we perceive and respond to the social world? Because metaphors can activate different kinds of social knowledge, they can influence how we interpret events (Landau, Meier, & Keefer, 2010). Consider just a few metaphors:

Her presentation bombed; everyone affiliated with her tried to run for cover.

He lifted the spirits of the audience; he received a warm reception.

Where is our relationship heading? Are we on the right track?

What you should notice first is that although you may not have heard any of those specific metaphors before, you can easily understand what is being communicated. In each of these examples, abstract concepts are being used to give a particular meaning to a concrete event. In the first sentence, people's knowledge of warfare is being used to structure our understanding of people's response to the contents of a talk. In the second example, both weight and temperature are used to guide our understanding of people's response to the contents of another talk. In the last example, the concept of a journey or travel is being applied to love and relationships.

Does such metaphor use have consequences for social judgment and behavior? New research is emerging that suggests this is so (Landau et al., 2010). Table 2.1 presents a selection of metaphors, which when primed, can influenced a number of different types of relevant social inferences and behavior. Let's just consider one example. In order to make the contamination metaphor available, Landau, Sullivan, and Greenberg (2009) had participants first read about the many airborne bacteria in the environment, which were described as either harmful to humans or not. Then, in a seemingly unrelated task about American domestic issues, statements relating to the United States were presented using the body metaphor ("After the Civil War, the United States experienced an unprecedented *growth spurt*") or without it ("After the Civil War, the United States experienced an unprecedented period of innovation"). In the third phase of the study, participants were asked to indicate their attitudes toward immigration. For those with a concern about "body contamination"—because they'd been told about how bacteria can harm humans—more negative attitudes toward immigration were expressed when the metaphor of the United States as a body had been made salient compared to when the United States had been described without this metaphor. So, how we talk—literally the pictures we paint with our words—can affect how we interpret and respond to the social world.

TABLE 2.1 Metaphors Can Affect Social Attitudes and Behavior

A variety of metaphors, when primed, have been shown to affect attitudes, memory, judgments, and physical perceptions.

METAPHOR PRIMING	EFFECT ON SOCIAL JUDGMENT
Nations are bodies (Landau, Sullivan & Greenberg, 2009)	Framing U.S. as body led to harsher attitudes toward immigration in those motivated to protect their body from contamination
Good is up; Bad is down (Crawford, Margolies, Drake, & Murphy, 2006)	Positive items presented in higher location and negative items in lower location recalled best
God is up (Chasteen, Burdzy, & Pratt, 2009)	Photos of people presented in a high (vs. low) position on screen were judged as having a stronger belief in God
Social exclusion is physical cold (Zhong & Leonardelli, 2008)	Recalling a time of social exclusion (vs. acceptance) resulted in the room being perceived as 5 degrees colder
Past is backward; Future is forward (Miles, Nind, & Macrae, 2010)	Backward postural sway was exhibited when thinking of the past and forward sway shown when thinking of the future

(*Source:* Based on research by Landau, Meier, & Keefer, 2010).



KEY POINTS

- A basic component of social cognition are **schemas**—mental frameworks developed through experience that, once formed, help us to organize and make sense of social information.
- Once formed schemas exert powerful effects on what we notice (attention), enter into memory (encoding), and later remember (retrieval). Individuals report remembering more information that is consistent with their schemas than information that is inconsistent with them, but in fact, inconsistent information, too, is strongly represented in memory.
- Schemas are often primed—activated by experiences, events, or stimuli. Once they are primed, the effects of the schemas tend to persist until they are somehow expressed in thought or behavior; such expression (known as **unpriming**) then reduces the likelihood they will influence thought or behavior.
- Schemas help us to process information, but they show a strong **perseverance effect** even in the face of disconfirming information, thus distorting our understanding of the social world.
- Schemas can also exert self-fulfilling effects, causing us to behave in ways that create confirmation of our expectancies.
- **Metaphors**—linguistic devices that relate an abstract concept to another dissimilar concept—can shape how we perceive and respond to the social world.

Automatic and Controlled Processing: Two Basic Modes of Social Thought

Social thought can occur in either of two distinctly different ways: in a systematic, logical, and highly effortful manner known as *controlled processing*, or in a fast, relatively effortless, and intuitive manner known as **automatic processing.** This distinction has been confirmed in literally hundreds of different studies and it is now recognized as an important aspect of social thought. But this doesn't mean that these two kinds of thought are totally independent; in fact, recent evidence suggests that automatic and controlled processing may often occur together, especially in situations involving some uncertainty (Sherman et al., 2008). Still, the distinction between them is important and worth us considering very carefully.

While a great deal of evidence supports the existence of these two different modes of social thought, perhaps the most convincing support is provided by the kind of social neuroscience research described briefly in Chapter 1—research that examines activity in the human brain as an individual processes social information. The findings of such research suggest that people actually possess two different neural systems for processing social information—one that operates in an automatic manner and another that operates in a systematic and controlled manner. Moreover, the operation of these two systems is reflected by activation in different regions of the brain. For instance, consider research on *evaluative reactions*—a very basic kind of social judgment relating to whether we like or dislike something (a person, idea, or object). Such evaluations can occur in two distinct ways: simple good–bad judgments that occur in a rapid and seemingly automatic manner (Phelps et al., 2001) or through more effortful thought in which we think carefully and logically, weighing all the relevant points fully and systematically (e.g., Duncan & Owen, 2000). The first kind of reaction seems to occur primarily in the amygdala, while the second seems to involve portions of the prefrontal cortex (especially the medial prefrontal cortex and ventrolateral prefrontal cortex (e.g., Cunningham, Johnson, Gatenby, Gore, & Banaji, 2003). In addition, as we'll note in a later discussion of the relationship between cognition and affect (between thought and emotions or moods), we also seem to possess two distinct brain systems for processing these types of information, with controlled processing (reasoning, logic) occurring primarily in the prefrontal cortex areas of the brain, and emotion-related, automatic reactions occurring mainly in the limbic system, structures deep inside the brain (e.g., Cohen, 2005).

Overall, the results of social neuroscience studies, as well as more traditional methods of social psychological research, suggest that the distinction between automatic and controlled processing is indeed real—and very important. We'll be illustrating this fact in many places throughout this book, but here, we'll try to clarify why it is so important by examining two specific issues relating to automatic processing: the effects of automatic processing on social behavior, and the benefits provided by such processing.

Automatic Processing and Automatic Social Behavior

automatic processing
This occurs when, after extensive experience with a task or type of information, we reach the stage where we can perform the task or process the information in a seemingly effortless, automatic, and nonconscious manner.

Once a concept is activated, it can exert important effects on social thought and behavior. Often, people act in ways that are consistent with their schemas, even if they do not intend to do so, and are unaware that they are acting in this manner. For example, in a well-known study by Bargh, Chen, and Burrows (1996), these researchers first activated either the schema for the trait of rudeness or the schema for the trait of politeness through priming. To do so, participants worked on unscrambling scrambled sentences containing words related either to rudeness (e.g., *bold*, *rude*, *impolitely*, *bluntly*) or words related to politeness (e.g., *cordially*, *patiently*, *polite*, *courteous*). People in a third (control) group unscrambled sentences containing words unrelated to either trait (e.g., *exercising*, *flawlessly*, *occasionally*, *rapidly*). After completing this task, participants in the study were asked to report back to the experimenter, who would give them additional tasks.

When they approached the experimenter, he or she was engaged in a conversation with another person (an accomplice). The experimenter continued this conversation, ignoring the participant. The major dependent measure was whether the participant interrupted the conversation in order to receive further instructions. The researchers predicted that people for whom the trait rudeness had been primed would be more likely to interrupt than those for whom the trait politeness had been primed, and this is precisely what happened. Further findings indicated that these effects occurred despite the fact that participants' ratings of the experimenter in terms of politeness did not differ across the three experimental conditions. Thus, these differences in behavior seemed to occur in a nonconscious, automatic manner.

In a second study, Bargh et al. (1996) either primed the stereotype for *elderly* (again through exposure to words related to this schema) or did not prime it. Then they timed the number of seconds it took participants to walk down a hallway at the end of the study. As predicted, those for whom the stereotype elderly had been primed actually walked slower! Together, the results of these and other studies (e.g., Dijksterhuis & Bargh, 2001) indicate that activating stereotypes or schemas can exert seemingly automatic effects on behavior—effects that occur in the absence of intention or conscious awareness. Clearly, then, automatic processing is an important aspect of social thought—one that can affect overt behavior.

But additional research suggests that the effects of automatic processing may be even more general than that of triggering particular forms of behavior. Once automatic processing is initiated (e.g., through priming), individuals may—again unconsciously—begin to prepare for future interactions with the people or groups who are the focus of this automatic processing. As suggested by Cesario, Plaks, and Higgins (2006), activating a schema may not merely trigger behaviors consistent with this schema; it may also activate behaviors that, in a sense, "get the people involved ready" to actually interact with others.

A study conducted by Cesario et al. (2006) clearly illustrates such effects. Participants were primed with photos of men labeled "GAY" or "STRAIGHT." These photos were shown so quickly that participants could not actually see the images; but as in many other studies, it was expected that the photos would prime (activate) schemas for these two groups. Then, in what seemed to be unrelated procedures, the computer on which the study was being conducted locked up, and participants were instructed to get the experimenter to help get it started. When the experimenter entered, he acted in a hostile manner. The key question was: would participants whose negative stereotype (schemas) of gays had been primed behave more hostilely than those whose stereotypes of heterosexuals had been primed? If so, this would be directly *contrary* to the stereotype of gays, which generally suggests that such people are passive and nonaggressive. However, it would be *consistent* with the view that priming this schema motivates individuals to prepare to interact with members of the people or group who are the focus of the schema—in this case, a group they do not like. Results offered clear support for this prediction: when interacting with the experimenter, participants did in fact show greater hostility if they had been primed with faces labeled "GAY" than with faces labeled "STRAIGHT." Remember: this activation was automatic because participants could not consciously report seeing these photos; they were presented for only 11 msec. The different predictions of these two views—(1) schemas trigger behaviors consistent with the schemas or (2) schemas trigger motivated preparation to interact with the people or groups who are the subject of the schemas—are summarized in Figure 2.8.

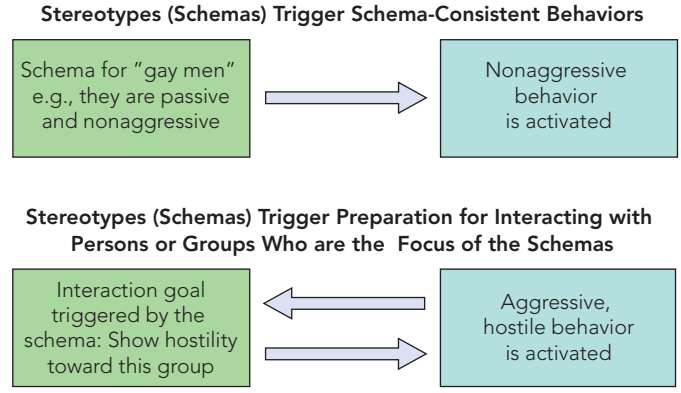


FIGURE 2.8 Automatic Processing Initiates Preparation for Future Interactions

Activation of schemas can trigger behaviors consistent with these cognitive frameworks. Recent research suggests that in addition, once activated, schemas may also trigger motivated efforts to prepare for interacting with the persons or groups who are the focus of these schemas. In the case of gay men, for instance, this enhances tendencies for heterosexuals to act in a hostile, aggressive manner. (Source: Based on suggestions by Cesario, Plaks, & Higgins, 2006).

The Benefits of Automatic Processing: Beyond Mere Efficiency

One kind of automatic processing with which most people are familiar occurs when we try to remember something (someone's name, a thought we previously had)—but don't succeed. When that happens, we often turn to doing something else while the search for the information we want goes on automatically, and without our conscious awareness. Often, this kind of memory search is successful, and the missing name or fact pops into mind. In such cases, we are dimly aware that *something* was happening, but can't really describe it. Research on this aspect of automatic processing confirms that we often attempt to deal with problems, and even complex decisions, while our attention is directed elsewhere (e.g., Dijksterhuis & Nordgren, 2007). Perhaps even more surprising, recent evidence indicates that sometimes it may be superior to careful, conscious thought in terms of making excellent decisions (Galdi, Arcuri, & Gawronski, 2008).

A clear illustration of these advantages is provided by research conducted by Dijksterhuis and van Olden (2006). These social psychologists asked students to look at various posters and indicate the one they liked most. In one condition (immediate decision), the posters were all shown on a computer screen simultaneously, and students made their decision immediately. In another condition (conscious thought), the posters were shown one at a time for 90 seconds, and after looking at them, the students were given paper and asked to list their thoughts and evaluations—to think carefully about the posters and their preferences for them. Finally, in a third condition (unconscious thought), participants worked on another task (solving anagrams) after seeing the posters, preventing them from consciously thinking about their preferences. Several minutes later, students indicated which poster they liked.

All the participants then received a surprise: they were given their favorite poster to take home. Three to five weeks later, they were phoned and asked how satisfied they were with the poster they had received and how much they would want (in Euros) if they sold their poster. The researchers predicted that participants would actually be most satisfied with their choice in the unconscious condition, where they made the choice without an opportunity to think consciously about it, and as you can see from Figure 2.9, this is precisely what happened. This suggests—surprisingly—that participants actually made better decisions, in terms of being satisfied with them, when they did so on "automatic" rather than when they had a chance to think about them carefully.

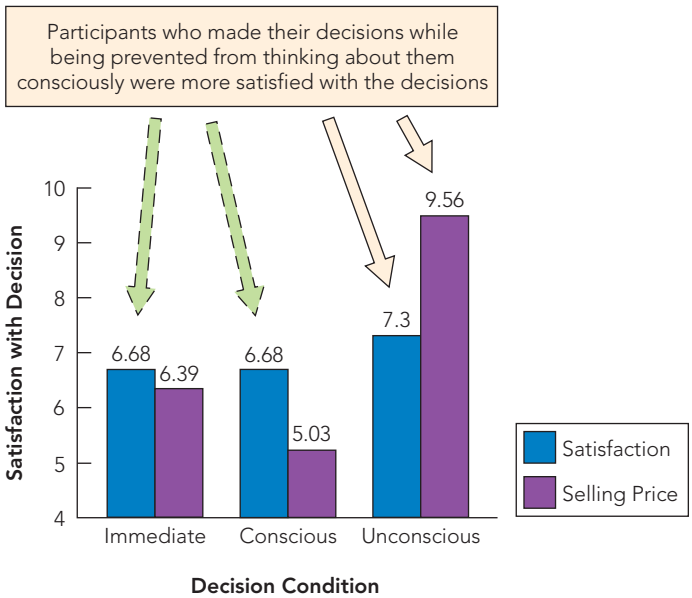


FIGURE 2.9 The Benefits of Automatic (Unconscious) Thought

Participants who were prevented from thinking consciously about their preferences for various posters (unconscious condition) were more satisfied with the choices they made than participants who could engage in careful, systematic thought (conscious) or participants who made their choice immediately after seeing the poster (immediate). These findings suggest that automatic processing offers more benefits than simply being quick and efficient. (Source: Based on data from Djiksterhuis & van Olden, 2006).

Why is this so? Perhaps because conscious thought has strict limits in terms of the amount of information it can handle, so when we think actively about decisions we may not be unable to take account of all available information. In contrast, unconscious, automatic thought has much greater capacity. Similarly, when we think about decisions consciously, we may fail to weight the various dimensions or elements accurately and thinking about these dimensions may get us confused about which were actually the most important. Unconscious, automatic processing may therefore reflect our real preferences more clearly. Whatever the precise reason, these findings, and those of many related studies (e.g., Ito, Chiao, Devine, Lorig, & Cacioppo, 2006), suggest that automatic processing offers important advantages beyond those of merely being quick and efficient. Certainly, there are real drawbacks to relying solely on conscious thought in making decisions, even though conscious thought is important in other ways, particularly in facilitating social

interaction (Baumeister & Masicampo, 2010). In our special feature "SOCIAL LIFE IN A CONNECTED WORLD: Dealing with Information Overload and Improving Choices," we consider the perils of relying on only conscious processes in environments that exceed our processing capacities.



SOCIAL LIFE *in a* CONNECTED WORLD

Dealing with Information Overload and Improving Choices

Get on almost any Internet site, and you're likely to be overloaded pretty quickly. As we have emphasized in this chapter, human beings are limited in the amount of information they can process and people routinely use heuristics to help them process all that incoming information. Information overload and the strategies people use to deal with it is similar to the problem of choice overload.

Barry Schwartz, in his 2004 book, *The Paradox of Choice,* talks about the negative consequences of having too many choices, a situation we can experience in both our online and bricks-and-mortar lives. Despite the negative reactions we can experience from having our choices restricted, as Schwartz points out, even when choosing a pair of jeans, the multiplicity of choices may give us a headache—both figuratively and literally! He isolates one factor as key: the whole idea of higher expectations. When we had only one type of jeans to pick from (Levi's 501s), and we had to break those ill-fitting jeans in, we could always blame the "world" for our discomfort. But when we have zillions of types of jeans to choose from, we can only blame ourselves if we don't end up with a perfect pair! After all, we made the choice, and there were so *many* to choose from!

While at first glance, it might seem wonderful that we have so many choices—for everything from health insurance plans to types of jeans to nail polish colors—but having so many choices can have a paralyzing effect. Not only that, even if the paralysis is overcome, we can end up less satisfied with our outcomes. What are some of the processes that lead to this negative effect? Well, the more options we have, the easier it is to imagine that another option than the one we chose would have been better than the one we actually did choose. Going back to the jeans example, even when we finally choose a pair of jeans, and it seems like an excellent choice, we still may be set up for an unexpected burden: long-term self-blame! We always feel that we could have done better, and so it is supremely easy to be disappointed when the options we have to choose from are abundant.

Turning to the online world, there is evidence that we might even be better off if we had fewer choices. Thaler and Sunstein (2008) take a stab at explaining how people might best deal with all the choices that Amazon, eBay, and other institutions offer us online. These researchers posit that we would be better off by limiting excessive choices by using "choice architecture," a method by which alternatives are crafted on the Web in order that people may more easily make better choices. Choice architecture simply means taking advantage of people's heuristic use in order to help them make the best choices. If we knew, for example, that people tend to choose the second option they see, these researchers suggest we should place the option that is likely to be best for most people in that position.

Take the potentially complicated issue of "school choice": only a tiny percentage of students actually switched schools when the choice was made available. In that situation, it was found that while many choices were offered, parents faced a very complex multistage process for getting their child transferred to another school.

In this case, parents used the "status quo" heuristic, as opposed to choosing a school that might be better equipped to help their child. Given that parents had to access a 100-page booklet with descriptions of 190 schools written by employees of the schools, where each school's positive features were given—most chose not to! Even if they had done so, the booklet did not include information on physical location, test scores, attendance rates, and racial composition, although that information was available on the district Website for those who searched around to find it. Thus, parents would have needed to combine very complex information from two sources in order to select a good school for their child. No wonder virtually every parent chose not to do so!

So school administrators tried a novel experiment to address this problem. In the past, low-income parents tended to put less weight than high-income parents on school quality. In effect, this allowed for higher-income parents to unwittingly "game the system." In their experiment, a random sample of parents received a list of schools giving the average test scores as well as acceptance rates at various schools for which any given student was actually eligible. With this newer simplified presentation of the crucial information, would low-income parents select better schools? Turns out that parents who received the information in a way that highlighted the crucial information in an accessible style did

(continued)

SOCIAL LIFE *in a* CONNECTED WORLD *(continued)*

place more emphasis on school quality and low-income parents' school choice decisions were similar to parents whose incomes were much higher. Thaler and Sunstein's (2008) research makes clear that choice-making by people of all backgrounds can be improved, leading them to better lives, if we allow simple forms of choice architecture to be utilized.

The problem of information overload and the resultant excessive choices to be made is a daunting one. In the online world, we are constantly marketed to. In a social media environment, we're faced with tons of choices. In interactions between ordinary people and government agencies, there is often the presence of complex materials. In general, information overload has the effect of narrowing people's thinking processes, just when they need to systematically evaluate far too many options. Understanding the heuristics people use when faced with complex information can help improve people's ability to cope with the many choices that must be made—and that's increasingly important in our overloaded "cyber-world."



KEY POINTS

- A large amount of evidence indicates that the distinction between automatic and controlled processing is a very basic one. In fact, different regions of the brain appear to be involved in these two types of processing, especially with respect to evaluations of various aspects of the social world.
- When schemas or other cognitive frameworks are activated (even without our conscious awareness of such activation), they strongly influence our behavior, triggering actions consistent with the frameworks and also preparing us to interact with the people or groups who are the focus of these schemas.
- **Automatic processing** is clearly quick and efficient; in addition, however, it may also sometimes offer other advantages too—such as decisions with which we are more satisfied.
- Having available too many choices can be paralyzing, and encourages dissatisfaction with the choices we do make.
- Use of "choice architecture"—where the best alternative for most people is strategically placed so that people who are automatically processing are more likely to select that option—can improve decision making and satisfaction with the outcomes.

Potential Sources of Error in Social Cognition: Why Total Rationality Is Rarer Than You Think

Human beings are definitely not computers, and our thinking is not simply based on rational self-interest as economists have long assumed (Akerlof & Shiller, 2009). The judgments people make systematically deviate in a number of ways from perfect rationality; this is true for critical decisions such as what career path to pursue or whom to marry, as well as financial decisions such as picking stocks to invest in or credit card use—our actions often reflect overconfidence and optimism (Gärling, Kirchler, Lewis, & van Raaij, 2009). While we can *imagine* being able to reason in a perfectly logical way, we know from our own experience that often we fall short of this goal. In our efforts to understand others and make sense out of the social world, we are subject to a wide range of tendencies that, together, can lead us into serious error. We now consider several of these "tilts" in social cognition. Before doing so, however, we should emphasize the following point: While these aspects of social thought do sometimes result in errors, they can also be adaptive. They often reduce the effort required for navigating the social world. As we saw with heuristic use—they supply us with tangible benefits as well as exacting important costs.

As we'll soon see, there are many different ways in which our social thought departs from rationality. To acquaint you with a wide range of these effects, we start with a basic

tendency that seems to occur in a wide range of situations and often produces important errors in our social thought: our tendency to be optimistic—often, overly so. After considering this far-reaching general tendency, we turn to several other ways in which social thought departs from rationality, ones that are also important but tend to occur in specific situations rather than generally like our tendency to be overly optimistic.

A Basic "Tilt" in Social Thought: Our Powerful Tendency to Be Overly Optimistic

If we were completely rational in the ways in which we think about the social world, we would simply gather information, process it, and then use it to make judgments and decisions. Instead, in many ways, most people tend to "see the world through rose-colored glasses," which is known as the **optimistic bias**–a powerful predisposition to overlook risks and expect things to turn out well. In fact, research findings indicate that most people believe that they are *more* likely than others to experience positive events, and *less* likely to experience negative events (Shepperd, Carroll, & Sweeny, 2008). Our strong leaning toward optimism can be seen in many specific judgments—most people believe that they are more likely than others to get a good job, have a happy marriage, and live to a ripe old age, but less likely to experience negative outcomes such as being fired, getting seriously ill, or getting divorced (Kruger & Burrus, 2004; Schwarzer, 1994).

Similarly, we often have greater confidence in our beliefs or judgments than is justified—an effect known as the **overconfidence barrier.** Vallone, Griffin, Lin, and Ross (1990) illustrated how overconfident people can be in their predictions about themselves by asking students to indicate early in the academic year whether they would perform a number of actions (e.g., drop a course, move on or off campus) and to indicate how confident they were in their predictions. The students were wrong a substantial proportion of the time, and even when they were 100 percent confident in their predictions they were wrong 15 percent of the time!

Ironically enough, people who are *least* competent in a domain are often the *most* likely to be overconfident of their judgments in that domain! Like many other types of judgments, we frequently have to assess our competence under conditions of uncertainty—where all the relevant information is not known. Consider just a few examples: have we picked the best health insurance plan to meet our future needs, are our retirement funds sufficiently diversified to weather even a rocky stock market, is our new kitchen design optimal, are the essays we write for class covering all the essential points on the topic? Caputo and Dunning (2005) have pointed out that one critical reason why we may be overly confident of our judgments and actions in all these cases is that we often are lacking critical information—that is, we do not know enough to know what we have missed. These researchers argue that for many tasks overconfidence stems from *errors of omission*. Suppose you were asked to come up with as many uses as possible for WD-40, an oil lubricant. You come up with what you think is an impressive list of 20 legitimate uses for it. Would you then see yourself as competent at this task? Based on the research conducted by Caputo and Dunning, people do confidently rate their abilities as high under these circumstances, but they should not because they have no way of knowing the other 1,980 legitimate uses for this product that they have missed! Indeed, when these researchers told their participants the possible solutions to their tasks that had been missed, people's confidence in their ability dropped and then more strongly correlated with objective measures of performance. So, one important reason we display overconfidence is that we lack the relevant feedback that would help moderate our confidence. As the cartoon in Figure 2.10 suggests, overconfidence may explain why entrepreneurs who start a new business believe that their chances of making it work are much higher than is actually true (Baron & Shane, 2007).

optimistic bias
Our predisposition to expect things to turn out well overall.

overconfidence barrier
The tendency to have more confidence in the accuracy of our own judgments than is reasonable.



FIGURE 2.10 Overconfidence in Action: Believing You'll Score Big Before You Have Started

As research findings (Baron & Shane, 2007) indicate, business entrepreneurs frequently express greater confidence in their likelihood of succeeding than the objective odds would warrant.

THE ROCKY PAST VERSUS THE GOLDEN FUTURE: OPTIMISM AT WORK Think back over your life. Did it have peaks—times when things were going great for you, and valleys—times when things were not good? Now, in contrast, try to imagine your future: How do you think it will unfold? If you are like most people, you may notice a difference in these descriptions. While most of us recognize that our past has been mixed in terms of "highs" and "lows," we tend to forecast a very rosy or golden future—one in which we will be quite happy and in which few negative events happen to us. In fact, research by Newby-Clark and Ross (2003) indicates that this tendency is so strong that it occurs even when people have just recalled negative episodes from their own pasts. What accounts for this difference? One possibility is that when we think about the past, we can recall failures, unpleasant events, and other disappointments, whereas these unexpected possibilities are not salient when we think about our future. When we think about the future, in contrast, we tend to concentrate on desirable goals, personal happiness, and doing things we have always wanted to do—such as traveling to exotic places. Since our thinking is dominated by these positive thoughts, we make highly optimistic predictions about the future, and tend to perceive it as indeed golden, at least in its promise or potential for us. In short, the optimistic bias seems to occur not just for specific tasks or situations, but for projections of our entire future as well.

Perhaps people also feel optimistic about the future—because it just feels good to do so! But, still, might there be hidden costs of being optimistic about ourselves and our future—particularly if we get there and find that optimism was misplaced? New research by Sweeny and Shepperd (2010) has addressed these questions. Students in a psychology class were asked to estimate the grade they would receive on their first exam and their emotional state was measured. Then, the students received their grade and their emotions were again measured. First of all, those students who were more optimistic about the grade they would receive reported more positive emotions, suggesting that optimism does feel good. But what happens when the students learned whether their optimism was warranted or not (i.e., they learned their exam grade)? For those optimistic students who overestimated their exam scores, when they learned their actual score, they felt much worse than the realists or pessimists who did not do so. The good news is, however, 24 hours later, the negative emotions the optimists felt had dissipated. This means that while being optimistic about our future outcomes can make us feel good, if the basis for it is disconfirmed, we may feel bad—but fortunately only temporarily!

WHEN OPTIMISM AFFECTS OUR ABILITY TO PLAN EFFECTIVELY Yet another illustration of optimism at work is the **planning fallacy**—our tendency to believe that we can get more done in a given period of time than we actually can, or that a given job will take less time than it really will. We can see this aspect of the optimistic bias in announced schedules for public works (e.g., new roads, airports, bridges, stadiums) that have no chance of being met. Individuals, too, adopt unrealistically optimistic schedules for their own work (see Figure 2.11). If you have ever estimated that a project would take you

planning fallacy
The tendency to make optimistic predictions concerning how long a given task will take for completion.

a certain amount of time but then found that it took considerably longer, you are already familiar with this effect, and with the planning fallacy.

Why do we (repeatedly) fall prey to this particular kind of optimism? According to Buehler et al. (1994), social psychologists who have studied this tendency in detail, several factors play a role. One is that when individuals make predictions about how long it will take them to complete a given task, they enter a *planning* or *narrative* mode of thought in which they focus primarily on the future and how they will perform the task. This, in turn, prevents them from looking backward in time and remembering how long similar tasks took them in the past. As a result, one important "reality check" that might help them avoid being overly optimistic is removed. In addition, when individuals *do* consider past experiences in which tasks took longer than expected, they tend to attribute such outcomes to factors outside their control. The result: they tend to overlook important potential obstacles that can't be easily foreseen when predicting how long a task will take, and fall prey to the planning fallacy. These predictions have been confirmed in several studies (e.g., Buehler et al., 1994), and they provide important insights into the origins of the tendency to make optimistic predictions about task completion.



FIGURE 2.11 The Planning Fallacy
The tendency to believe that the plans we construct are doable, that we can accomplish more than we actually can in a given period of time, or that nothing will interfere with the achievement of our goals reflects the planning fallacy in action. Few projects are actually completed as originally planned, or on schedule!

These cognitive factors are not the entire story, though. Additional findings suggest that another factor, *motivation* to complete a task, also plays an important role in the planning fallacy. When predicting what will happen, individuals often guess that what will happen is what they *want* to happen (Johnson & Sherman, 1990). In cases where they are strongly motivated to complete a task, people make overoptimistic predictions about when they will attain this desired state of affairs (Buehler, Griffin, & MacDonald, 1997). It appears, then, that our estimates of when we will complete a task are indeed influenced by our hopes and desires: we want to finish early or on time, so we predict that we will.

Are some people more prone to the planning fallacy than others? As we just discussed, when people are focused on the goal of completing a task, rather than the steps involved in doing so, they are likely to make overly optimistic predictions for how much time it will take to do so. Weick and Guinote (2010) proposed that people in powerful positions are more likely to fall prey to the planning fallacy because they are focused on the goal of getting the task done, whereas people who occupy less powerful positions are more likely to be focused on the *how* or the steps needed to be taken to get the job done. These researchers tested this idea by having some participants think about an episode in their past when they occupied a position of relative power, or an episode in which they were in a position of relative powerlessness. Subsequently, both groups of participants were asked to format a document using software that was complicated, but before actually doing so they were asked to estimate how long it would take them to do

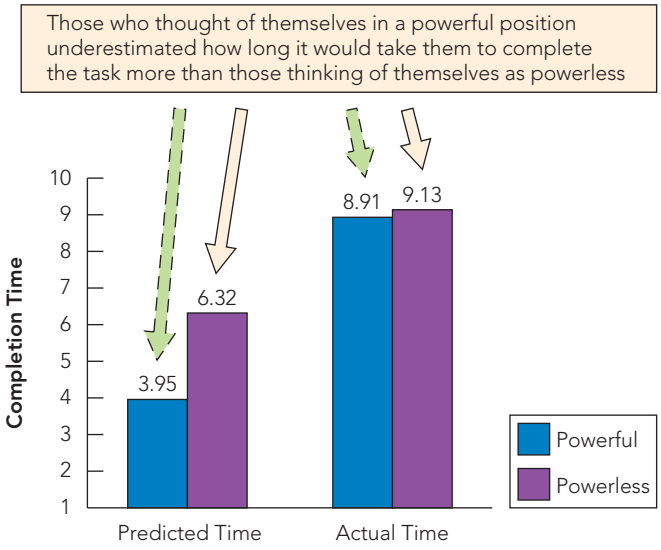


FIGURE 2.12 Power and the Planning Fallacy

Both powerful and powerless people seriously underestimated how long it would take them to complete a complex word processing task, but those who thought of themselves occupying a powerful position mispredicted the time that would be needed most. These results are consistent with the idea that power leads us to focus too narrowly on task completion, rather than the steps involved in getting there, which can lead us to seriously underestimate how long it will take us to finish a task. (Source: Based on research by Weick & Guinote, 2010).

so. As shown in Figure 2.12, both groups of participants showed the planning fallacy—that is, both groups seriously underestimated the number of minutes they would need to complete the editing task. However, as the researchers predicted, although there was no difference in actual performance time, those who first thought of themselves as occupying a position of power underestimated how long it would take them much more than did participants who thought of themselves as occupying a position of powerlessness. These results are consistent with the idea that power leads us to focus too narrowly on task completion, rather than the steps involved in getting there, which can lead us to seriously underestimate how long it will take to finish tasks.

Situation-Specific Sources of Error in Social Cognition: Counterfactual Thinking and Magical Thinking

The optimistic bias is very general in nature; as we've seen, it can be found in a wide range of social situations. Other important forms of bias in our social thought are more restricted in the sense that they tend to occur only in certain kinds of situations. We now examine two of these—*counterfactual thinking* and what is sometimes termed *magical thinking*.

COUNTERFACTUAL THINKING: IMAGINING "WHAT MIGHT HAVE BEEN" Suppose that you take an important exam; when you receive your score, it is a C−, a much lower grade than you had hoped. What thoughts will enter your mind as you consider your grade? If you are like most people, you may quickly begin to imagine "what might have been"—receiving a higher grade—along with thoughts about how you could have obtained that better outcome. "If only I had studied more, or come to class more often," you may think to yourself. And then, perhaps you may begin to formulate plans for actually *doing* better on the next test.

Such thoughts about "what might have been"—known in social psychology as **counterfactual thinking**—occur in a wide range of situations, not just ones in which we experience disappointments. For instance, suppose you read an article in the newspaper about someone who left work at the normal time and was injured in an automobile accident in which another driver ran a stop sign. Certainly, you would feel sympathy for this person and would probably recommend some form of compensation. But now imagine the same story with a slight difference: the same person was injured in the same kind of accident, but in this case, he had left work early to run an errand. Since the accident is the same, you should rationally feel the same amount of sympathy for the victim. But in fact, you may not because given that he left work earlier than usual, it is easy to imagine him *not* being in the accident. Or, suppose he took an unusual route home instead of his normal one. Would that make a difference in the sympathy you would feel? Research indicates that the answer is yes—emotional responses differ depending on how easy it is to mentally undo the circumstances that preceded it. Because it is easier to undo in our minds taking the unusual route than the normal one, sympathy for the accident will also differ. In other words, counterfactual thoughts about what might have happened instead of what did happen can influence your sympathy—as well as your recommendations concerning compensation for the victim (e.g., Miller & McFarland, 1987). This difference in the intensity of the sympathy evoked has been observed even for highly tragic events, including cases of rape and the loss of a child in an auto accident (Branscombe,

counterfactual thinking
The tendency to imagine other outcomes in a situation than the ones that actually occurred ("What might have been").

Owen, Garstka, & Coleman, 1996; Davis, Lehman, Wortman, Silver, & Thompson, 1995; Wolf, 2010).

Counterfactual thoughts seem to occur automatically in many situations—we simply can't help imagining that things might have turned out differently. To overcome these automatic tendencies, therefore, we must try to correct for their influence, and this requires both active processing in which we suppress the counterfactual thoughts or discount them. Consistent with this idea, studies have demonstrated that anything that reduces our information-processing capacity actually *strengthens* the impact of counterfactual thoughts on our judgments and behavior (Goldinger, Kleider, Azuma, & Beike, 2003). Together, this research indicates that counterfactual thinking—imagining what did not actually happen—can influence our social thought.

When counterfactual thinking does occur, a wide range of effects can follow—some of which are beneficial and some of which are costly to the people involved (Kray, Galinsky, & Wong, 2006; Nario-Redmond & Branscombe, 1996). Depending on its focus, imagining counterfactuals for outcomes we receive can yield either boosts to, or reductions in, our current moods. If individuals imagine *upward counterfactuals*, comparing their current outcomes with more favorable ones than they experienced, the result may be strong feelings of dissatisfaction or envy, especially when people do not feel capable of obtaining better outcomes in the future (Sanna, 1997). Olympic athletes who win a silver medal but who can easily imagine winning a gold one experience such reactions (Medvec, Madey, & Gilovich, 1995). Alternatively, if individuals compare their current outcomes with less favorable ones—it might have been worse—they may experience positive feelings of satisfaction or hopefulness. Such reactions have been found among Olympic athletes who win bronze medals, and who can easily imagine what it would be like to have not won any medal whatsoever. In sum, engaging in counterfactual thought can strongly influence current affective states, and willingness to gamble on obtaining those outcomes in the future (Petrocelli & Sherman, 2010).

In addition, it appears that we often use counterfactual thinking to mitigate the bitterness of disappointments. After tragic events such as the death of a loved one, people often find solace in thinking: "Nothing more could be done; the death was inevitable." In other words, they adjust their view concerning the inevitability of the death so as to make it seem more certain and therefore unavoidable. In contrast, if they have different counterfactual thoughts—"If only the illness had been diagnosed sooner . . ." or "If only we had gotten him to the hospital quicker . . ."—their suffering may be increased. So by assuming that negative events or disappointments were inevitable, it tends to make these events more bearable (Tykocinski, 2001).

Finally, we should note that counterfactual thinking can sometimes help us to perform better—to do a better job at various tasks. Why? Because by imagining how we might have done better, we may come up with improved strategies and ways of using our effort more effectively. So, sometimes—for instance, when we expect to repeat various tasks—engaging in counterfactual thought can enhance performance on important tasks (Kray et al., 2006). Our tendency to think not only about what is, but also about what might have been, therefore, can have far-reaching effects on many aspects of our social thought and social behavior.

MAGICAL THINKING, TERROR MANAGEMENT, AND BELIEF IN THE SUPERNATURAL

Please answer truthfully:

> *If you are in class and don't want the professor to call on you, do you try to avoid thinking about being called on?*
>
> *If you were given an opportunity to buy travel insurance, would you feel you were "tempting fate" and inviting calamity by not purchasing it?*
>
> *If someone offered you a piece of chocolate shaped like a cockroach—would you eat it?*



FIGURE 2.13 Magical Thinking: An Example
Would you eat the candy shown here? Many people would not, even though they realize that the shape of the candy has nothing to do with its taste. This illustrates the law of similarity—one aspect of what social psychologists term magical thinking.

On the basis of purely rational considerations, you know that your answers should be "no," "no," and "yes." But are those the answers you actually gave? Probably not. In fact, research findings indicate that human beings are quite susceptible to what has been termed **magical thinking** (Rozin & Nemeroff, 1990). Such thinking makes assumptions that don't hold up to rational scrutiny but that are compelling nonetheless (Risen & Gilovich, 2007). One principle of such magical thinking assumes that one's thoughts can influence the physical world in a manner not governed by the laws of physics; if you think about being called on by your professor, it does not change the probability that you actually will be! Likewise, simply sticking pins in a doll and thinking about it as hurting your enemy does not mean such "voodoo" really can result in harm to another person. But, based on the *law of similarity*, which suggests that things that resemble one another share basic properties, it might be easy to think that sticking a doll that looks like an enemy can cause the same kind of harm to the real person. For the same reason, people won't eat a chocolate shaped like a cockroach even though they know, rationally, that its shape has nothing to do with its taste (see Figure 2.13). People also seem to believe that they are "buying peace of mind" when they purchase insurance; that is, not only will they be covered if something does go wrong, but that the very act of buying the insurance will ensure it does not go wrong! Research indicates that by turning down an insurance opportunity, people believe they are "tempting fate" and increasing the likelihood that disaster will strike (Tykocinski, 2008).

Surprising as it may seem, our thinking about many situations is frequently influenced by such magical thinking. So, what is the basis of such seemingly nonrational thinking? Some theorists have suggested that because human beings are uniquely aware of the fact that we will certainly die, this, in turn, causes us to engage in what is known as **terror management**—efforts to come to terms with this certainty and its unsettling implications (Greenberg et al., 2003). One kind of thinking that helps is belief in the supernatural—powers outside our understanding and control—that can influence our lives. Recent research indicates that when we are reminded of our own mortality, such beliefs are strengthened (Norenzayan & Hansen, 2006). In short, when we come face to face with the certainty of our own deaths, we try to manage the strong reactions this produces, and one way of doing this is to engage in thinking that is largely outside of what we consider to be rational thought.

So, the next time you are tempted to make fun of someone's superstitious belief (e.g., fear of the number 13 or of a black cat crossing one's path), don't be too quick to laugh: Your own thinking is almost certainly *not* totally free from the kind of "magical" (i.e., nonrational) assumptions that seem to underlie a considerable portion of our social thought.

magical thinking
Thinking involving assumptions that don't hold up to rational scrutiny—for example, the belief that things that resemble one another share fundamental properties.

terror management
Our efforts to come to terms with certainty of our own death and its unsettling implications.



KEY POINTS

- Social thought departs from rationality in a number of ways. People show a strong optimistic bias, expecting that we are more likely than others to experience positive outcomes but less likely than others to experience negative ones.
- In addition, people tend to exhibit **overconfidence** in their predictions, and those who have the least competence in a domain are most likely to be overly confident of their judgments in that domain. This seems to be due to *errors of omission,* where we lack comparison information that would help moderate our confidence.
- People make more **optimistic** judgments about their future than their past. Optimism that is not born out in reality can result in negative emotions.
- People make overly optimistic predictions about how long it will take them to complete a given task, an effect known as the **planning fallacy.** This occurs repeatedly both because we fail to consider obstacles we may encounter when predicting how long a task will take and because we are motivated to complete a task so fail to consider all the time-consuming steps necessary to do so.
- In many situations, individuals imagine "what might have been"—they engage in **counterfactual thinking.** Such thought can affect our sympathy for people who have experienced negative outcomes. But upward counterfactuals can also motivate us to perform better in the future in hope of avoiding the outcome that did occur.
- There are important limits on our ability to think rationally about the social world. One involves magical thinking—assuming our thoughts can influence the physical world or that our actions (e.g., not buying insurance) may "tempt fate" and increase the likelihood of negative events. Based on similarity of two objects, we seem to believe that the properties of one can pass to the other.
- One form of such thinking—belief in the supernatural—stems, at least in part, from **terror management**—our efforts to cope with the knowledge that we will die. Reminders of our own mortality strengthen supernatural beliefs.

Affect and Cognition: How Feelings Shape Thought and Thought Shapes Feelings

Think of a time in your own life when you were in a very good mood—something good had happened and you were feeling very happy. Now, in contrast, remember a time when you were in a very bad mood—something negative had occurred and you were feeling down and blue. Was your thinking about the world different at these two times? In other words, did you remember different kinds of events or experiences, reason differently, and perhaps think about other people in contrasting ways? In all likelihood you did, because a large body of research findings indicate that there is a continuous and complex interplay between *affect*—our current moods or emotions—and *cognition*—various aspects of the ways in which we think, process, store, remember, and use information (e.g., Forgas, 2000; Isen & Labroo, 2003). We don't use the word *interplay* lightly because, in fact, existing evidence strongly suggests that the relationship between affect and cognition is very much a two-way street: Our emotions and moods strongly influence several aspects of cognition, and cognition, in turn, exerts strong effects on our emotions and moods (e.g., Baron, 2008; McDonald & Hirt, 1997; Seta, Hayes, & Seta, 1994). We now take a closer look at the nature of these effects.

The Influence of Affect on Cognition

First, and perhaps most obviously, our current moods can influence our perceptions of the world around us. When we are in a good mood (experiencing positive affect), we tend to perceive almost everything—situations, other people, ideas, even new inventions—in

more positive terms than we do when we are in a negative mood (Blanchette & Richards, 2010; Clore, Schwarz, & Conway, 1993). Indeed, this effect is so strong and so pervasive that we are even more likely to judge statements as true if we encounter them while in a positive mood than if we read or hear them while in a neutral or negative mood (Garcia-Marques, Mackie, Claypool, & Garcia-Marques, 2004). Positive moods can also encourage people to feel they understand the world better (e.g., Hicks, Cicero, Trent, Burton, & King, 2010). When these researchers presented stimuli that have inherent ambiguity—such Zen koans as "If a placebo has an effect, is it any less real than the real thing?" or abstract art pictures—to participants, those in positive moods consistently reported greater understanding had been derived from the stimuli, particularly among participants that had first reported that they tend to use heuristics when making judgments (e.g., agree with statements such as "I rely on my intuitive impressions").

Such effects have important practical implications. For instance, consider their impact on job interviews—a context in which interviewers meet many people for the first time. A growing body of evidence indicates that even experienced interviewers cannot avoid being influenced by their current moods: They assign higher ratings to the people they interview when they are in a good mood than when they are in a bad mood (e.g., Baron, 1993a; Robbins & DeNisi, 1994). While positive moods can increase our confidence about our interpretation given to actions performed by other people, they can also result in less accuracy (Forgas, Vargas, & Laham, 2005).

Another way in which affect influences cognition involves its impact on memory. Here, two different, but related, kinds of effects seem to occur. One is known as **mood congruence effects.** This refers to the fact that current moods strongly determine which information in a given situation is noticed and entered into memory. In other words, current moods serve as a kind of filter, permitting primarily information consistent with these moods to enter into long-term storage. Second, affect also influences what specific information is *retrieved* from memory, an effect known as **mood dependent memory** (e.g., Baddeley, 1990; Eich, 1995). When experiencing a particular mood, individuals are more likely to remember information they acquired in the past while in a similar mood than information they acquired while in a different mood. Current moods, in other words, serve as a kind of *retrieval cue*, prompting recall of information consistent with these moods. Here's an illustration of the difference between these two effects. Suppose that you meet two people for the first time. You meet one when you are in a very good mood but meet the other one when you are in a very bad mood (e.g., you just learned that you did poorly on an important exam). Because of mood congruence effects, you will probably notice and store in memory mainly positive information about the first person, but you are more likely to notice and store in memory mainly negative information about the second person. Your mood when you meet these people determines what you notice and remember about them.

Now, imagine that at a later time, you are in a good mood. Which person comes to mind? Probably, the one you met while in a similar (good) mood. Here, your current mood serves to trigger memories of information you acquired (and stored in memory) when you were in a similar mood in the past. Together, mood congruence and mood-dependent memory strongly influence the information we store in memory. Since this is the information we can later remember, the impact of affect on memory has important implications for many aspects of social thought and social behavior. Figure 2.14 summarizes these points concerning mood and memory.

mood congruence effects
The fact that we are more likely to store or remember positive information when in a positive mood and negative information when in a negative mood.

mood dependent memory
The fact that what we remember while in a given mood may be determined, in part, by what we learned when previously in that mood.

Our current moods also influence another important component of cognition: creativity. The results of several studies suggest that being in a happy mood can increase creativity—perhaps because being in a happy mood activates a wider range of ideas or associations than being in a negative mood, and creativity consists, in part, of combining such associations into new patterns (Estrada, Isen, & Young, 1995; Isen, 2000). A recent meta-analysis combining all the studies investigating the relationship between mood and creativity (Baas, De Dreu, & Nijstad, 2008) indicates that positive moods facilitate creativity most when they are relatively high in arousal (e.g., happiness) rather than low in arousal (e.g., relaxation).

A third way in which affect influences cognition involves the tendency to engage in *heuristic processing*, thinking that relies heavily on mental "shortcuts" (heuristics) and knowledge acquired through past experience. This, in turn, has important implications for decision making and problem solving—activities we all perform frequently. Research findings indicate that people experiencing positive affect are more likely than people experiencing negative affect to engage in heuristic thought (i.e., to rely on previously acquired "rules of thumb" and previously gathered information) in dealing with current problems or decisions (Mackie & Worth, 1989; Park & Banaji, 2000; Wegner & Petty, 1994). If these are applicable to the new situation, they can be helpful. If not, they can get in the way of both effective decision making and performance.

Finally, we should mention that our current moods often influence our interpretations of the motives behind people's behavior. Positive affect tends to promote attributions of positive motives, while negative affect tends to encourage attributions of negative motives (Forgas, 2000). As we note in Chapter 3, our thoughts about the cause of others' behavior play an important role in many situations, so this is another way in which the interplay between affect and cognition can have important effects.

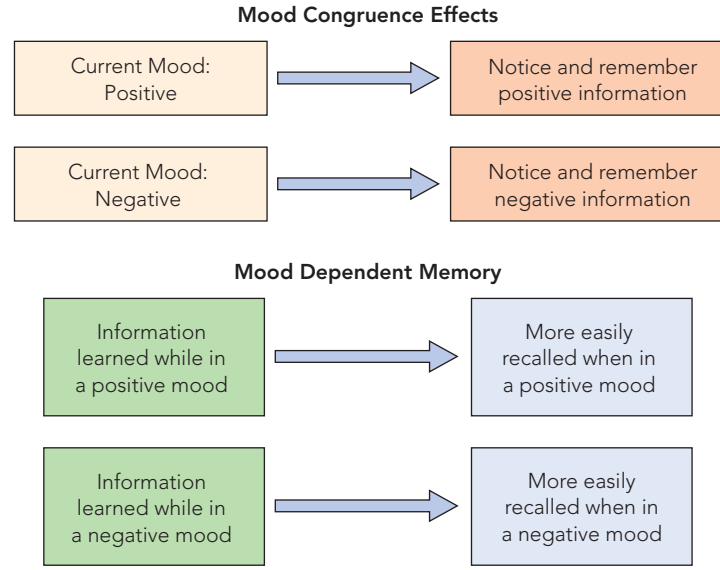


FIGURE 2.14 The Effects of Mood on Memory
Our moods influence what we remember through two mechanisms: mood congruence effects, which refer to the fact that we are more likely to store or remember information consistent with our current mood, and mood dependent memory, which refers to the fact that we tend to remember information consistent with our current moods.

The Influence of Cognition on Affect

Most research on the relationship between affect and cognition has focused on how feelings influence thought. However, there is also strong evidence for the reverse: the impact of cognition on affect. One aspect of this relationship is described in what is known as the *two-factor theory* of emotion (Schachter, 1964). This theory suggests that often, we don't know our own feelings or attitudes directly. Rather, since these internal reactions are often somewhat ambiguous, we infer their nature from the external world—from the kinds of situations in which we experience these reactions. For example, if we experience increased arousal in the presence of an attractive person, we may conclude that we are in love. In contrast, if we experience increased arousal after being cut off in traffic by another driver, we may conclude that what we feel is anger.

A second way in which cognition can influence emotions is by activating schemas containing a strong affective component. For example, if we categorize an individual as belonging to a group different than our own, we may experience a different emotional response than if we categorized that same individual as a member of our own group. Let's consider a case of watching a person receive a seemingly painful needle injection in the hand. When the picture was of an African hand, Caucasian participants exhibited lower empathic reactions as indicated by reduced brain activity in the pain areas of the brain relative to when the picture of the hand receiving the injection was also Caucasian (Avenanti, Sirigu, & Aglioti, 2010). The same results—in reverse—were observed for participants of African descent; greater empathic pain reactions in the brain were observed when the hand was Black compared to when it was White. These results indicate that how we think about others—and who we think those others are—tells us how we feel about such people, and whether we "feel" their pain or not. But, do we always know how we will feel about the suffering of others? For detailed information on this important issue, please see our special section, "EMOTIONS AND SOCIAL COGNITION: Why We Can't Always Predict Our Responses to Tragedy."

EMOTIONS *and* SOCIAL COGNITION

Why We Can't Always Predict Our Responses to Tragedy

Would you feel worse if you learned that one person was killed in a forest fire, or if you learned that 1,000 people were? Most people believe that they would feel worse upon learning about the large-scale tragedy compared to the smaller-scale one. Yet, much research indicates that our **affective forecasts**—predictions about how we would feel about an event we have not experienced—are often inaccurate (Dunn & Laham, 2006). To the extent that our cognition (affective forecasts) is based on a different way of processing information compared to actual emotional experience, these two types of responses—forecasting and experiencing—should differ. Because rational cognition is responsive to abstract symbols, including numbers, forecasting should vary depending on the scale of the tragedy being considered. Emotions, in contrast, which are based on concrete images and immediate experiences, may be relatively insensitive to the actual numbers of people killed, or more generally the scope of a tragedy.

To test this idea—that affective forecasting will be responsive to numbers, but that people who are actually experiencing the images from a tragedy will show an "emotional flatline" as the death toll increases, Dunn and Ashton-James (2008) conducted a number of studies. In one experiment, one group of participants was placed in the "experiencer role"; they were given a news article about a deadly forest fire in Spain and were asked to report their actual emotions while reading about the tragedy. Another group of participants was placed in the "forecaster role" and they were simply asked to predict how they would feel "if they read about a deadly forest fire in Spain." The scope of the tragedy of the fire was also varied. Some participants were told that five people had been killed, while other participants were told that 10,000 people had been killed by the fire.

Did the size of the tragedy affect how bad participants actually reported feeling in the experience condition or they expected to feel in the forecasting condition? Yes, the size of the tragedy did affect how forecasters expected to feel, but the number of people killed in the fire did *not* affect how people actually reported feeling. Not only did forecasters overestimate how bad they would feel overall, but they believed they would be responsive to the magnitude of the tragedy whereas those who were actually exposed to the tragic loss information showed a "flatline" response and did not differentiate their emotional response according to numbers.

In a subsequent study, these researchers brought the tragedy closer to home—the victims were members of their own group. Students were told that either 15 or 500 American college students had been killed in the war in Iraq, and pictures of the sort shown in Figure 2.15 were presented to



FIGURE 2.15 Emotional Responses to the Tragegy of One or Many
People who are asked to forecast how they would feel about the tragic deaths of others believed they would feel worse as the number of people killed increased. However, people who were actually given the detailed information to read or view felt about the same regardless of how many people had died. This research is consistent with the idea that rational information processing, which occurs in forecasting, differs from actual emotional experience.

the *experiencers* on a website prepared for the study. The *forecaster* participants were not shown the actual pictures or website, but were asked to imagine how they would feel if they viewed one of the website versions. Again, participants who were only forecasting how they would feel overestimated their negative affect compared to the experiencers, and the forecasters were sensitive to the number of deaths while the experiencers were not.

Forecasting affective responses to tragedy may not only lead to inaccuracies in general (overestimates of how distressed people will be). Forecasting appears to also result in specific errors: expecting greater mobilization on the part of others as the scope of the tragedy increases, although those who are actually exposed to and consuming images of the tragedy do not respond differentially according to the numbers of people who have suffered.

A third way in which our thoughts can influence our affective states involves our efforts to regulate our own emotions and feelings. This topic has important practical implications, so we'll examine it carefully.

affective forecasts
Predictions about how we would feel about events we have not actually experienced.

COGNITION AND THE REGULATION OF AFFECTIVE STATES Learning to regulate our emotions is an important task; negative events and outcomes are an unavoidable part of life, so learning to cope with the negative feelings these events generate is crucial for personal adjustment—and for good social relations with others. Among the most important techniques we use for regulating our moods and emotions are ones involving cognitive mechanisms. In other words, we use our thoughts to regulate our feelings. Many techniques for accomplishing this goal exist, but here, we'll consider one that is especially common—giving in to temptation as a means of improving our current mood.

When we feel "down" or distressed, we often engage in activities that we know might be bad for us in the long run, but that make us feel better, at least temporarily (e.g., engage in some "retail therapy" by going shopping, eat fattening snacks, drink alcohol; see Figure 2.16). These actions make us feel better, but we know full well that they have an important "downside." Why, then, do we choose to do them? In the past it was assumed that people engage in such actions because the emotional distress we are experiencing reduces either our capacity or motivation to control our impulses to do things that are enjoyable but potentially bad for us. However, Tice et al. (2000) argue that cognitive factors in fact play a role in such behavior; we yield to such temptations because it helps us deal with strong negative feelings.

To test this prediction, Tice et al. (2000) conducted a study in which participants were first put into a good or bad mood (by reading stories in which they either saved a child's life or ran a red light and caused the death of a child). Then, participants were either told that their moods could change over time or their moods were "frozen" and could not change much. Participants then were led to believe they would work on an intelligence test on which they would receive feedback. Before doing the test, though, they would have a 15-minute practice session to prepare for it. The experimenter then left them in a room containing materials for practicing for the test *and* distracters—other tasks on which they could work. For half the participants these tasks were attractive and tempting (e.g., a challenging puzzle, a video game, popular magazines). For the others, they were less attractive (a preschool-level plastic puzzle, out-of-date technical journals). The main question was this: would people in a bad mood spend more of the practice time than people in a good mood playing with the distracters (procrastinating)? More importantly, would this occur only in the condition where participants believed they could change their own moods? After all, there would be no use in playing with the distracters if participants believed that their moods were "frozen" and could not be altered. Tice et al. predicted that people in a bad mood would procrastinate more, but only when they



FIGURE 2.16 Consciously Regulating Our Negative Moods
When people are feeling down, many engage in activities designed to make them feel better—they go shopping, consume alcohol, and so on. Research findings suggest that engaging in such actions is the result of conscious strategy for regulating our emotions.

believed doing so would enhance their moods—and the results offered clear support for the prediction. These findings indicate that the tendency to yield to temptation is a conscious choice, not a simple lapse in the ability to control our own impulses.

Affect and Cognition: Social Neuroscience Evidence for Two Separate Systems

So far we have argued that affect and cognition are intimately linked, and in fact, existing evidence suggests that this is certainly the case. However, we should also note that recent findings using neuroscience techniques (e.g., scanning of human brains as individuals perform various activities) indicate that actually two distinct systems for processing social information may exist within the human brain (e.g., Cohen, 2005). One system is concerned with what might be termed "reason"—logical thought—whereas the other deals primarily with affect or emotion. These two systems, although distinct in certain respects, interact in many ways during problem solving, decision making, and other important forms of cognition. For instance, consider research employing what is known as an "ultimatum" paradigm.

In such research, two people are told that they can divide a given sum (e.g., $10) between them. One person can suggest an initial division and the second can accept or reject it. Since any division provides the second person with positive payoffs, total rationality (and classic economic theory) suggests that acceptance of any division offered is the most rational (and best) course of action. In fact, however, most

people reject divisions that give them less than $3, and many reject divisions that offer them less than $5. Magnetic resonance imaging (MRI) scans of the brains of people performing this task reveal that when they receive offers they view as unfair, brain regions related both to reasoning (e.g., the dorsolateral prefontal cortex) and to emotion (e.g., the limbic system) are active. However, the greater the amount of activity in the emotion-processing regions, the greater the likelihood that individuals will reject the offers—and act in ways that are, in a sense, contrary to their own economic interests (e.g., Sanfey, Rilling, Aronson, Nystrum, & Cohen, 2003). These findings, and those of many other studies, provide concrete evidence for the existence of two distinct systems (reason and emotion) that interact in complex ways during decision making and other cognitive processes (e.g., Gabaix & Laibson, 2006; Naqvi, Shiv, & Bechara, 2006).

Additional research indicates that the neural system for emotion tends to be impulsive, preferring immediate rewards, whereas the system for reason is more forward-looking and accepting of delays that ultimately yield larger rewards. For instance, when offered the choice between an immediate gain (a $15 Amazon.com gift now) and a larger one in 2 weeks (a $20 gift voucher), increased activity occurs in both emotion-related and reason-processing regions of the brain. The immediate option, however, induces greater activity in the emotion-related areas (e.g., the limbic system; McClure, Laibson, Loewenstein, & Cohen, 2004).

Overall, then, evidence from research using modern techniques for scanning brain activity during cognitive processes suggests that affect plays a fundamental role in human thought, and that if we wish to fully understand the complex ways in which we think about the social world and our place in it, we must take this fact into careful account because certain aspects of our thought can also influence our feelings. Affect and cognition are not one-way streets; they are a divided highway, with the potential of one influencing the other.



KEY POINTS

- Affect influences cognition in several ways. Our current moods can cause us to react positively or negatively to new stimuli, including other people, the extent to which we think systematically or heuristically, and can influence memory through **mood dependent memory** and **mood congruence effects**.
- When we are in a positive mood, we tend to think heuristically to a greater extent than when we are a negative mood. Specifically, we show increased reliance on stereotypes and other mental shortcuts.
- Cognition influences affect through our interpretation of emotion-provoking events and through the activation of schemas containing a strong affective component. Brain activity reflective of empathy in response to pain experienced by another person depends on how we categorize the other person.
- **Affective forecasts**—predictions about how we would feel about an event we have not experienced—are often inaccurate because cognition and affect are based in different systems. Those in a forecasting role are sensitive to the numbers of people harmed, whereas those in an experience role are not differentially responsive to the magnitude of the tragedy.
- We employ several cognitive techniques to regulate our emotions or feelings. For instance, when distressed, we can consciously choose to engage in activities that, while damaging in the long run, make us feel better in the short run.
- Research in social neuroscience indicates that we may actually possess two distinct systems for processing social information—one concerned with logical thought and the other with affect or emotion.



SUMMARY *and* REVIEW

- Because we have limited cognitive capacity, we often attempt to reduce the effort we expend on **social cognition**—how we think about other people and the social world. Given our limited capacity to process information, we often experience **information overload.** To cope with this, we make use of **heuristics**—simple rules of thumb—for making decisions in a quick and relatively effortless manner. One such heuristic is **representativeness,** which suggests that the more similar an individual is to typical members of a given group, the more likely she or he is to belong to that group. When using the representativeness heuristic, people tend to ignore base rates—frequencies of events or patterns in the total population. Another heuristic is **availability,** which suggests that the easier it is to bring information to mind, the greater its impact on subsequent decisions or judgments. Use of availability can lead us astray to the extent that vivid events are easier to bring to mind, but as not necessarily more frequent in occurrence. A third heuristic is **anchoring and adjustment,** which leads us to use a number or value as a starting point from which we then make adjustments. These adjustments may not be sufficient to reflect actual social reality, perhaps because once we attain a plausible value, we stop the process. A fourth heuristic, status quo, leads us to favor "old" over "new."

- One basic component of social cognition is **schemas—**mental frameworks developed through experience that, once formed, help us to organize social information. Once-formed schemas exert powerful effects on what we notice (attention), enter into memory (encoding), and later remember (retrieval). Individuals report remembering more information consistent with their schemas than information inconsistent with them, but in fact, inconsistent information too is strongly represented in memory. Schemas are often *primed*—activated by experiences, events, or stimuli. Once they are primed, the effects of the schemas tend to persist until they are somehow expressed in thought or behavior; such expression (known as **unpriming**) then reduces their effects. Schemas help us to process information, but they often persist even in the face of disconfirming information. Schemas can also exert self-fulfilling effects, causing us to behave in ways that confirm them. **Metaphors,** which relate an abstract concept to another dissimilar one, can shape how we respond to the social world.

- A large amount of evidence indicates that the distinction between automatic and controlled processing is a very basic one. In fact, different regions of the brain appear to be involved in these two types of processing, especially with respect to evaluations of various aspects of the social world. When schemas or other cognitive frameworks are activated (even without our conscious awareness of such activation), they can influence our behavior, triggering actions consistent with the frameworks and also preparing us to interact with the people or groups who are the focus of these schemas. Automatic processing is quick and efficient; in addition, however, it may also sometimes offer other advantages too—such as increased satisfaction with decisions. Decisions we must make under **conditions of uncertainty** can be improved with "choice architecture," which involves identifying the heuristics people use and placing the options in the order and format where most people are likely to select the option that will benefit them.

- People show a strong **optimistic bias,** expecting positive events and outcomes and fewer negatives in many contexts. In addition, people tend to be overconfident in their judgments and predictions about themselves. This occurs because people make errors of omission; they lack the comparison information that would allow them to know what factors they have not considered. One example of our optimism at work is the **planning fallacy**—our tendency to believe that a task will take less time than it really will. In many situations, individuals imagine "what might have been"—they engage in **counterfactual thinking.** Such thought can affect our sympathy for people who have experienced negative outcomes. Counterfactual thinking seems to occur automatically in many situations, and adding cognitive load strengthens its impact on judgments.

- There are important limits on our ability to think rationally about the social world. One involves **magical thinking**—thinking based on assumptions that don't hold up to rational scrutiny. For instance, we may believe that if two objects are in contact, properties can pass from one to the other. One form of such thinking—belief in the supernatural—stems, at least in part, from **terror management**—our efforts to cope with the knowledge that we will die.

- Affect influences cognition in several ways. Our current moods influence our perceptions of the world around us, the extent to which we think systematically or heuristically, and influence memory through mood-congruence effects and mood-dependent memory. Affect can also influence creativity and our interpretations of others' behavior. Cognition influences affect through our interpretation of emotion-provoking events and through the activation of schemas containing a strong affective component. In addition, we employ several cognitive techniques to regulate our emotions or feelings (e.g., consciously giving in to temptation to reduce negative feelings). Although affect and cognition are closely related, social neuroscience research indicates that they involve distinct systems within the brain. People make **affective forecasts**—predictions about how they would feel about an event they have not experienced—using the cognitive system, but respond with the emotional system when confronted with those events.

KEY TERMS

affect (p. 37)
affective forecasts (p. 63)
anchoring and adjustment heuristic (p. 41)
automatic processing (p. 48)
availability heuristic (p. 40)
counterfactual thinking (p. 56)
conditions of uncertainty (p. 38)
heuristics (p. 37)
information overload (p. 38)
magical thinking (p. 58)
metaphor (p. 46)
mood congruence effects (p. 60)
mood dependent memory (p. 60)
optimistic bias (p. 53)
overconfidence barrier (p. 53)
perseverance effect (p. 46)
planning fallacy (p. 54)
priming (p. 45)
prototype (p. 38)
representativeness heuristic (p. 38)
schemas (p. 44)
social cognition (p. 36)
terror management (p. 58)
unpriming (p. 45)